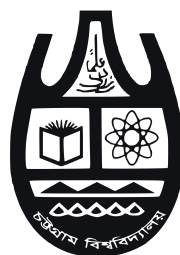


A Neuro-Symbolic Multi-Agent Framework for Pre-Release Audience Response Generation in Bangla Cinema

A Verifier-in-the-Loop Approach



Sabbir Hossain
Student ID: 21701082
Session: 2020-2021

Supervisor: Prof. Dr. Mohammad Shahadat Hossain

Department of Computer Science and Engineering
UNIVERSITY OF CHITTAGONG

This dissertation is submitted for the degree of
Bachelor of Science in Engineering (B.Sc. Engg.) in CSE

Report Code:

University of Chittagong

Department of Computer Science and Engineering

8th Semester B.Sc. Engineering Examination 2024

Course No.: CSE 800

Title: A Neuro-Symbolic Multi-Agent Framework for Pre-Release Audience Response Generation in Bangla Cinema

Subtitle: A Verifier-in-the-Loop Approach

Report Code:

University of Chittagong

Department of Computer Science and Engineering

8th Semester B.Sc. Engineering Examination 2024

Course No.: CSE 800

Title: A Neuro-Symbolic Multi-Agent Framework for Pre-Release Audience Response Generation in Bangla Cinema

Subtitle: A Verifier-in-the-Loop Approach

Student Name: Sabbir Hossain

Student ID: 21701082

Session: 2020-2021

Hall: Shaheed Abdur Rab

Signature of Student:

Submission Date: 03 September, 2026

SUPERVISOR APPROVAL PAGE

Title of the Thesis: A Neuro-Symbolic Multi-Agent Framework for Pre-Release Audience Response Generation in Bangla Cinema: A Verifier-in-the-Loop Approach

Document Type: Bachelor of Science in Engineering Thesis

Degree Program: Bachelor of Science in Engineering (B.Sc. Engg.) in Computer Science and Engineering

Institution: Department of Computer Science and Engineering, University of Chittagong

Student Name: Sabbir Hossain

Student ID: 21701082

Session: 2020–2021

Date of Submission: 3rd September, 2026

This Bachelor of Science in Engineering thesis has been reviewed by the supervisor and is hereby:

☐ Approved ☐ Not Approved ☐ Approved with Revisions

Evaluation Criteria (to be filled in by the supervisor)

Evaluation Criterion	Yes	No	Partly
Maintained regular communication	☐	☐	☐
Maintained professionalism	☐	☐	☐
Addressed the comments provided by the supervisor	☐	☐	☐
Checked using plagiarism and AI-content detection tools	☐	☐	☐
The thesis is approved for submission	☐	☐	☐

Comments, if any:

Signature of the Supervisor

Prof. Dr. Mohammad Shahadat Hossain

Department of Computer Science and Engineering
University of Chittagong

Declaration

I hereby declare that this thesis, entitled *A Neuro-Symbolic Multi-Agent Framework for Pre-Release Audience Response Generation in Bangla Cinema: A Verifier-in-the-Loop Approach*, is my own work, except where the work of others has been appropriately acknowledged and referenced.

I further declare that this thesis has not been submitted, either in whole or in part, to this or any other university or institution for the award of any degree or qualification. All sources of information used in this thesis have been duly acknowledged. This dissertation contains fewer than 65,000 words including appendices, bibliography, footnotes, tables and equations and has fewer than 150 figures.

Sabbir Hossain

Student ID: 21701082

Session: 2020–2021

Department of Computer Science and Engineering

University of Chittagong

Chattogram 4331, Bangladesh

Email: shk.sabbir.7@gmail.com

I dedicate this thesis to my beloved parents, *Md. Shahidul Islam* and *Dalia Islam*; to *Neha Khan*, someone very dear to me; and to my beloved sister, *Sanjida Islam Afra*, whose love, prayers, encouragement, and unwavering support have been a constant source of strength throughout my journey and have made this achievement possible.

Acknowledgements

First and foremost, I offer my deepest gratitude to Almighty Allah for His infinite mercy, guidance, and blessings in every aspect of my life. I am especially grateful to Him for granting me the strength, patience, and perseverance necessary to complete my bachelor's thesis successfully.

I would like to express my profound gratitude to my supervisor, **Prof. Dr. Mohammad Shahadat Hossain**, for his invaluable guidance, insightful suggestions, constructive feedback, and constant encouragement throughout this research. His expertise and thoughtful supervision played a vital role in shaping this work and bringing it to completion.

I am sincerely grateful to **Arfin**, **Tonmoy**, and **Monir** for generously contributing their time and effort to the human-evaluation component of this research. Their cooperation and assistance were invaluable to the successful completion of the evaluation.

Finally, I extend my heartfelt appreciation to everyone who supported, encouraged, or assisted me, directly or indirectly, throughout this academic journey.

Abstract

Audience commentary does not exist before viewers encounter a film, so a pre-release system cannot be validated simply by producing fluent review-like text. This thesis asks whether a language model can generate short Bangla (Bengali) cinema responses at a requested level of engagement specificity while keeping generation guidance separate from final evaluation. The study uses the Bangladeshi variety written in Bengali script. An audit of 5,000 reviews left a frozen analysis set of 4,625 rows and showed that the initial grouping largely identified two source blocks in the corpus, reaching 93.3% source-block accuracy. Because the data did not support natural audience groups, the target was redefined as two levels on an engagement-specificity continuum. Two blinded annotators recognized this distinction in 39/50 and 42/50 length-matched cases, against a 0.25 chance rate. The generation workflow retrieves examples from one data partition and uses an in-loop verifier to guide at most three drafts. A second verifier, trained on disjoint data, sees only the completed outputs. The main experiment produced 5,400 responses from 90 held-out plots, two requested levels, ten conditions, and three paired seeds. All nine alternatives improved the isolated verifier’s target probability over zero-shot generation. Neural gating with symbolic feedback produced the largest observed difference, +0.2570 (95% paired-bootstrap CI [0.2151, 0.2987]), although its advantage over neural-only gating remains exploratory. During 147 same-case revisions, both verifiers improved, but the in-loop verifier improved more, widening their gap by 0.1828. On a balanced subset of 100 items, three blinded native-Bangla annotators recovered the requested level with 0.9133 accuracy (95% CI [0.8667, 0.9567]). The findings demonstrate control of a human-recognizable property of generated text, not prediction of real audiences or film reception.

Keywords: Bangla NLP; controllable text generation; verifier-in-the-loop; neuro-symbolic AI; multi-agent systems; human evaluation.

Table of contents

List of figures	xii
-----------------	-----

List of tables	xiii
----------------	------

1	Introduction	1
1.1	Background	1
1.2	Motivation	2
1.3	Problem Statement	2
1.4	Research Aim and Objectives	3
1.5	Research Questions	3
1.6	Research Design Overview	4
1.7	Scope and Limitations	5
1.8	Contributions	6
1.9	Thesis Structure	7
1.10	Chapter Summary	7
2	Related Work	8
2.1	Synthetic Audiences and Response Control	8
2.2	Revision with Feedback	9
2.3	Retrieval-Augmented Generation	10
2.4	Neural and Symbolic Verification	11
2.5	Bounded Multi-Agent Workflows	11
2.6	Verifier Reliability and Proxy Gaming	12
2.7	Bangla NLP for This Study	12
2.7.1	Classification Backbones	12
2.7.2	Generation and Evaluation Resources	13
2.8	Human Evaluation of Controlled Generation	13
2.9	Research Gap	14

2.10 Chapter Summary	15
3 Data and Construct Validation	16
3.1 Study Design	16
3.2 Data Sources	18
3.2.1 Bangla Review Corpus	18
3.2.2 Plot Corpus	18
3.3 Cleaning and Quality Control	18
3.4 Frozen Data Split	19
3.5 Corpus Source Diagnostics	20
3.6 Developing the Engagement-Specificity Axis	21
3.7 Human Validation	23
3.7.1 Failed Ordinal-Rating Study	23
3.7.2 Comparative-Judgment Validation	24
3.8 Operational Definition	25
3.9 Ethics, Reproducibility, and Method Limits	25
3.10 Chapter Summary	26
4 Verifier Development and Isolation	27
4.1 Verifier Design and Data Separation	27
4.2 Backbone Results and Construction Circularity	28
4.2.1 Construction Circularity	28
4.3 Final Verifier Configurations	30
4.3.1 Verifier-A	30
4.3.2 Verifier-B	30
4.4 Probability Calibration	31
4.5 Isolation, Reproducibility, and Limits	32
4.6 Chapter Summary	33
5 Bounded Neuro-Symbolic Workflow	34
5.1 Roles and State	34
5.1.1 Functional Roles	34
5.1.2 Shared State and Trace	35
5.2 Generation and Retrieval	36
5.2.1 Bounded Generation Loop	36
5.2.2 Retrieval and Prompt Construction	37
5.3 Neural Gating and Symbolic Diagnosis	37

5.3.1	Separation of Decision and Feedback	37
5.3.2	Acceptance Threshold and Retry Limit	38
5.4	Experimental Conditions	40
5.4.1	Matched-Budget Blind Resampling	40
5.5	Execution, Reproducibility, and Limits	41
5.6	Chapter Summary	42
6	Results and Analysis	43
6.1	Design and Statistical Tests	43
6.1.1	Run Completeness	44
6.2	Results Across Conditions	44
6.2.1	Differences Between Levels	44
6.2.2	Paired Comparisons and the Neural–Symbolic Contrast	45
6.3	Retry Dynamics and Proxy Divergence	46
6.4	Blinded Human Evaluation	48
6.4.1	Human and Verifier Decisions on the Same Items	49
6.5	Length-Matched Sensitivity Analysis	49
6.6	Diversity and Distributional Diagnostics	50
6.7	Qualitative Error Analysis	51
6.8	Scope of the Comparisons	52
6.9	Chapter Summary	52
7	Discussion and Limitations	54
7.1	Main Finding	54
7.1.1	Engagement Specificity as a Continuum	55
7.2	What the Control Mechanisms Add	55
7.2.1	Why Verifier Separation Matters	56
7.3	Human and Automatic Evidence	56
7.4	Practical Implications	57
7.5	Limitations	57
7.5.1	Construct and Measurement	57
7.5.2	Internal and Statistical Validity	57
7.5.3	External Validity, Human Evaluation, and Ethics	58
7.6	Chapter Summary	58
8	Conclusion and Future Work	59
8.1	Conclusion	59

Table of contents	xi
8.2 Future Work	59
References	61

List of figures

3.1	Study design and data-isolation boundaries.	17
3.2	K-selection diagnostics for Region A and the Region-B negative control. . .	22
4.1	Backbone performance and the frozen-LaBSE baseline on dev-82.	29
4.2	Confusion matrices on dev-82.	31
5.1	Bounded generation workflow and post-hoc Verifier-B evaluation.	35
5.2	Development quality–cost frontier and forced-attempt score trajectories. . .	39
6.1	Paired Verifier-B effects relative to zero-shot; whiskers show 95% bootstrap confidence intervals.	47
6.2	Verifier trajectories and same-case gap changes among continuing failures. .	47
6.3	Length and feature-space diagnostics across conditions.	51
6.4	Rule-selected Bangla responses.	53

List of tables

2.1	How the present study differs from adjacent work	15
3.1	Review-corpus audit and cleaning	19
3.2	Frozen review partitions and permitted uses	20
3.3	Source-region differences in the raw workbook	21
3.4	Construct evidence and Region-B negative control	23
3.5	Results of the two construct-validation studies	24
3.6	The two engagement-specificity levels	25
4.1	Data available to each verifier	28
4.2	Model and baseline performance on dev-82	29
4.3	Roles and data boundaries of the final verifiers	31
4.4	Calibration results on dev-82	32
5.1	Roles, inputs, outputs, and model access	34
5.2	Neural and symbolic signals on development outputs	38
5.3	Development operating points for threshold selection	39
5.4	The ten experimental conditions	40
6.1	Verifier-B outcomes and generation cost by condition	45
6.2	Paired effects relative to zero-shot	46
6.3	Human recognition of the requested level	48
6.4	Human and verifier decisions on 100 blinded items	49
6.5	Outcomes in the length-matched sensitivity slice	50
6.6	Selection basis for the reported examples, seed 42	52

Chapter 1

Introduction

This chapter defines the problem: controlling engagement specificity in short Bangla cinema responses before audience reviews are available. It introduces the research questions, design, scope, and the separation between generation-time and outcome verification.

1.1 Background

Audience commentary becomes available only after viewers encounter a film. This thesis asks whether a language model can generate short, plot-based Bangla responses before such commentary exists. Large language models (LLMs) can produce plausible responses from a synopsis, and recent work has evaluated this capability specifically for movie-review generation [55]. Fluent review-like text, however, establishes neither control of a requested characteristic nor credible prediction of real audience reception. Research on synthetic audiences similarly warns that fluent outputs may reproduce social and representational biases, respond inconsistently to prompt wording, hallucinate user characteristics, and encourage anthropomorphic interpretations beyond the available evidence [35].

Producing a fluent Bangla comment is only part of the task. The requested level of engagement specificity must be recognizable, controllable, and evaluated by a scorer that did not guide the generation process. Classifier-guided generation provides a precedent for conditioning text on a target attribute [64]. In this study, the target construct is developed for Bangla, revision is limited to three attempts, and the final evaluator is kept outside the generation process.

The study focuses on Bangla text written in Bengali script and on short, informal cinema commentary in the Bangladeshi research context. BanglaBERT provides a pretrained encoder for Bangla language understanding [7], while Language-Agnostic BERT Sentence Embedding (LaBSE) provides multilingual sentence representations [18]. The primary re-

view resource is the 5,000-row *Raw Bangla Movie Review Comment Dataset for Sentiment Analysis and Natural Language Processing* [26]. Its reviews are short and sentiment-labelled, but the dataset contains no movie-title field, audience demographic information, or row-level record of the original collection source. It can therefore support a bounded study of textual response patterns and controllability, but not film-linked audience modelling, demographic audience segmentation, or claims that generated responses predict what actual viewers will say.

1.2 Motivation

Prompting alone does not demonstrate control. The target distinction may be poorly defined, or a model may satisfy it through a shortcut such as response length. Repeated revision can also raise the score of a visible evaluator by exploiting its weaknesses. Intrinsic self-correction is unreliable without dependable external feedback [27, 30].

Controlled-generation methods use discriminators to influence an attribute [64], retrieval-augmented generation supplies contextual evidence [38], and generate–critique–revise procedures attempt to improve initial outputs. Synthetic-audience research examines risks of generated human proxies [35]. It remains unclear whether a response distinction derived from Bangla reviews can be controlled during generation and evaluated by a verifier that was not used to optimize the text. The present study combines the existing mechanisms to examine that question; it does not treat the mechanisms themselves as new. The generated comments are therefore treated as experimental texts, not as predictions of audience behaviour. Verifier-A guides revision, whereas Verifier-B scores the completed outputs and is never shown to the generator. This separation allows the analysis to detect disagreement between the two verifiers. Such disagreement matters because optimization against a visible evaluator can exploit its weaknesses, as evaluator stress tests have shown [3].

1.3 Problem Statement

The review corpus does not contain engagement-specificity labels. A usable distinction must therefore be derived without confusing it with sentiment, text length, or differences between corpus sources. The generator must then produce the requested level under a fixed retry policy. Finally, any gain reported by the in-loop scorer must be checked by an evaluator trained on disjoint data and excluded from prompt development, acceptance decisions, and retry routing.

Corpus analysis did not support natural audience categories. Instead, it supported an operational **engagement-specificity continuum**. At Level 0, a response is general, formulaic, or only weakly connected to a particular film detail. At Level 1, it engages with a specific event, character, relationship, or narrative element. These levels constitute a reproducible cut through a continuum. They are neither discovered demographic groups nor evidence that the source reviews contain naturally separated clusters.

The central question is whether a task-trained verifier can improve control of this distinction when placed inside a generate–verify–refine workflow. The comparison includes zero-shot generation and the other registered controls, while Verifier-B remains outside the loop. The study also separates the use of symbolic rules for acceptance from their use as revision feedback.

1.4 Research Aim and Objectives

The aim of this thesis is to design and evaluate an auditable verifier-in-the-loop framework for controlling engagement specificity in short Bangla cinema responses while keeping generation guidance separate from final outcome evaluation. The specific objectives are to:

1. Recover an operational engagement-specificity distinction from the Bangla review corpus and determine whether it remains stable and human-recognizable after testing corpus-region, sentiment, and length confounds.
2. Implement the bounded verifier-guided workflow and quantify how its registered prompting, retrieval, critique, judging, and resampling conditions change target-level controllability relative to zero-shot generation.
3. Determine how symbolic information functions when used for acceptance gating and when used as diagnostic feedback alongside neural verification.
4. Test whether iterative optimization against the in-loop Verifier-A produces measurable divergence from the isolated outcome Verifier-B.

1.5 Research Questions

This study addresses the following research questions:

- **RQ1:** Can a meaningful response distinction be recovered from unlabeled Bangla reviews and validated as stable and human-recognizable?

- **RQ2:** To what extent do verifier-guided generation and the registered prompting, retrieval, self-critique, external-judge, and resampling controls improve target-level controllability over zero-shot generation in Bangla?
- **RQ3:** What role does symbolic information play when used for acceptance gating and for diagnostic feedback within verifier-guided Bangla response generation?
- **RQ4:** Does iterative optimization against Verifier-A produce measurable divergence from the independent Verifier-B?

In RQ1, stability refers to reproducibility of the two-level cut; it is not evidence of natural audience categories. The confirmatory tests for RQ2 compare each of the nine registered conditions with zero-shot generation. Comparisons among the active conditions are descriptive or exploratory. RQ3 treats acceptance rules and diagnostic feedback as different uses of symbolic information. RQ4 uses Verifier-B as an independent outcome measure, not as human ground truth. The four questions address control and evaluation of generated text. Audience prediction, demographic segmentation, and commercial forecasting fall outside their scope.

1.6 Research Design Overview

The review corpus supports construct development, retrieval, and verifier training; 120 Bangla Wikipedia film synopses supply separate generation stimuli. They cannot form film–review pairs because reviews lack movie titles.

Cleaning and near-duplicate control leave 4,625 reviews for the frozen split. The 300-item Gold partition is reserved for evaluation. R1 supplies retrieval examples and training data for Verifier-A; R2 is used only to train Verifier-B. The initial full-corpus partition identified corpus region with 93.3% accuracy, which points to a source or style difference rather than audience personas. In Region A, the two-way cut was reproducible, but the silhouette was 0.053, the gap statistic selected no value of K , and HDBSCAN labelled every review as noise. These results do not support natural clusters.

Length remained a possible shortcut: a length-only classifier achieved AUC 0.6764. The surface and residual profiles are therefore reported as descriptions, not as independent validation. The first Gold-300 instrument, which used ordinal ratings, failed its reliability gate (Krippendorff’s $\alpha = 0.4970$); no downstream validity test was performed. A second instrument used length-matched comparisons. The two annotators found the different item in 39/50 and 42/50 intrusion sets and chose the more engagement-specific response in 34/40

directional pairs. Chance performance was 0.25 and 0.50, respectively, while a length heuristic scored 0.16 on the intrusion task. The distinction is perceptible to the annotators, but it remains a cut through a continuum.

Generation proceeds through four functional roles. The Researcher retrieves same-level examples from the R1 index [38], and the Writer produces a response to the film plot. The Critic applies Verifier-A and the symbolic checks; the Reflector turns a failed check into revision guidance. These roles form one controlled workflow rather than four autonomous agents. Neural scores determine acceptance in the main condition, while symbolic rules identify specific problems in the draft. Each case permits at most three Writer attempts, and no LLM is fine-tuned.

Verifier-A uses LaBSE features to reproduce labels that were also derived from LaBSE space. Its near-perfect label accuracy is therefore partly circular: it shows that the cut can be learned within the same representation family, but it does not provide independent human validation. That evidence comes from the comparative annotation task. Verifier-B instead uses BanglaBERT, is trained on R2, and remains unavailable until final scoring.

Thirty development plots are used for prompt, threshold, and retry-policy work. The main experiment uses 90 held-out plots, two levels, ten conditions, and three paired seeds, producing 5,400 frozen outputs. Zero-shot and nine registered alternatives include a realized model-token-budget-matched blind resampling control. Verifier-B is applied only after generation. Seeds are paired sensitivity blocks, not independent replications.

All nine registered alternatives improved Verifier-B target probability relative to zero-shot. Neural gating with symbolic diagnostic feedback produced the largest observed registered zero-shot contrast, +0.2570 with a 95% paired-bootstrap confidence interval of [0.2151, 0.2987], but this does not establish superiority over other active conditions. A post-hoc comparison with neural-only gating remains exploratory. On a frozen balanced 100-item subset, three blinded native-Bangla annotators recovered the requested level with pooled accuracy 0.9133 and a 95% item-bootstrap interval of [0.8667, 0.9567]. Same-case revisions among continuing failed cases also widened the Verifier-A–Verifier-B gap in the neural-gated loops, consistent with proxy overoptimization but not proof of reduced human quality.

1.7 Scope and Limitations

The study is limited to short Bangla cinema responses at two levels of engagement specificity. It tests control of those levels and the separation of the two verifiers; it does not forecast audience opinion.

Source reviews lack film, audience, venue, and row-level provenance. The corpus has no separated natural clusters; its levels remain a continuum cut. Although construct-validation items were length-matched, generated length remained informative about requested level, precluding a length-neutral control claim. The generation study covers Bangla only; the incomplete English mirror supports no cross-lingual conclusion. Its three seeds are sensitivity blocks, not independent replications.

The comparative instrument followed the failed ordinal instrument and departed from the pipeline’s prescribed remedy. Although preregistered before judgment, it was designed after the first result; this remains a limitation, and a third instrument was prohibited. The blind-resampling comparator matches realized total tokens under the same generator rather than measured FLOPs, latency, hardware utilization, or the functional role of calls.

The human evaluation covers a balanced sample of 100 from the 5,400 outputs. It shows that readers can recover the requested level in that sample; it does not rank the ten conditions or assess naturalness, factual support, plot faithfulness, persuasiveness, or audience preference. Verifier-B is an outcome proxy, not human ground truth, and temperature scaling did not establish an improvement in its calibration. A widening gap between the verifiers is evidence of proxy divergence, not evidence that people would judge the revised text as worse. The direct comparison of neural-plus-symbolic feedback with neural-only gating was post-hoc and remains exploratory. The results are also tied to short cinema responses, Wikipedia film plots, the tested prompts, and the principal generator family. Chapter 7 discusses these limits in detail.

1.8 Contributions

This thesis makes six contributions:

1. A verifier-isolation protocol separates generation control from outcome evaluation and measures divergence on revisions of the same case.
2. The neuro-symbolic workflow records every attempt and reports both logical calls and token use across a frozen ten-condition experiment.
3. The construct-validation process rejects the initial persona interpretation, records the failed ordinal instrument, and retains only the human-recognizable engagement-specificity continuum supported by the length-matched comparison task.
4. The circularity analysis explains why reproducing LaBSE-derived labels with a LaBSE probe is not independent evidence of human validity.

5. A blinded human study tests whether readers can recover the requested level from a frozen, balanced sample of generated Bangla responses.
6. The results distinguish symbolic acceptance decisions from symbolic diagnostic feedback. Symbolic-only acceptance was weak, while the added value of symbolic feedback beside neural gating remains exploratory.

These contributions address controllability, verifier isolation, and the evidence needed to justify the target construct. They do not establish audience prediction or general superiority of a multi-role system. All the materials are available in [25].

1.9 Thesis Structure

The thesis is organized into eight chapters, with each chapter addressing a distinct part of the research:

- **Chapter 1** introduces the problem, research questions, study design, scope, and contributions.
- **Chapter 2** reviews the relevant work on response control, revision, retrieval, verification, Bangla NLP, and human evaluation.
- **Chapter 3** develops and validates the engagement-specificity axis from the review corpus.
- **Chapter 4** develops the two verifiers and defines their isolation.
- **Chapter 5** presents the bounded generation workflow and experimental conditions.
- **Chapter 6** reports the computational and human-evaluation results.
- **Chapter 7** interprets the findings and examines their limitations and implications.
- **Chapter 8** closes the thesis with the conclusion and future work.

1.10 Chapter Summary

The thesis asks whether short Bangla cinema responses can express a requested engagement-specificity level without allowing the generation-time verifier to judge its own outputs. Chapter 2 places that question in the relevant literature.

Chapter 2

Related Work

This chapter reviews the work that informs the study: synthetic audiences, controlled generation, self-correction, retrieval, neural and symbolic verification, multi-agent workflows, Bangla NLP, and human evaluation. The review focuses on what each area contributes to controlled Bangla response generation and where its evidence stops. The final section uses those limits to define the research gap.

2.1 Synthetic Audiences and Response Control

Language models are increasingly used to approximate audiences, users, or personas before direct human evidence is available. One model can produce many responses under different conditions, but plausible language is not evidence of behavioural validity. A generated response may resemble an intended persona without predicting how any person or population would respond. A recent systematic review identifies hallucination, bias, prompt sensitivity, limited ecological validity, and anthropomorphic interpretation as recurring problems in synthetic-audience studies [35]. The risk is greatest when generated text is presented as evidence about a community rather than as a hypothesis for later human research.

Mixture-of-Personas illustrates one approach to population-conditioned generation: latent or synthesized persona representations are used to steer a common language model toward diverse response distributions [10]. SimAB extends the synthetic-audience idea to persona-conditioned agents for rapid web A/B-test evaluation and compares simulated decisions with historical outcomes [50]. Both studies show how explicit conditioning can broaden the range of generated responses. Their aim is population simulation or behavioural prediction. This thesis asks a narrower question: whether a requested level of engagement specificity can be expressed and recognized in Bangla comments.

Controlled generation provides a second relevant tradition. FUDGE modifies token probabilities using a learned future discriminator while leaving the base generator unchanged [64]. This shows that an external classifier can steer generation without fine-tuning the generator. FUDGE applies its discriminator during decoding. The system studied here instead scores a completed draft and may request another attempt. FUDGE also has no separate outcome evaluator for testing whether optimization has exploited the steering model. It is a methodological precursor, not the same architecture.

Movie-review generation is the closest application domain. Sands et al. evaluate reviews written by LLMs and distinguish fluent persona-conditioned text from evidence of authentic audience response [55]. Generated comments in this thesis are treated as experimental texts, not as predicted reviews of a film.

2.2 Revision with Feedback

Iterative refinement assumes that a language model can inspect a draft, find its mistakes, and produce a better answer. The evidence is conditional. Huang et al. find that intrinsic self-correction—revision without reliable external feedback—does not consistently improve the reasoning tasks they examine and can alter correct answers [27]. Kamoi et al. similarly distinguish intrinsic feedback from correction supported by an oracle, tool, environment, or separately trained model, finding stronger evidence when feedback is externally grounded [30]. These findings show that generation ability does not guarantee reliable self-evaluation.

Self-Refine operationalizes an intrinsic generate–feedback–revise cycle using the same model [40]. Reflexion stores verbal feedback in memory and can draw that feedback from internal or external signals depending on the task [57]. Both systems make intermediate critique explicit. Their results do not show that self-generated criticism is always accurate, or that reflection rather than the additional model calls causes the improvement.

External verification provides an alternative. Cobbe et al. train a verifier to rank sampled completions and report that verifier-based selection improves performance relative to additional generator fine-tuning [14]. The result supports an independently trained selection signal, but it concerns ranking a fixed candidate pool rather than revising one draft. The experiment in this thesis includes both verifier-guided revision and blind resampling.

Role placement is also consequential. Recent evidence shows that the effect of explicit error feedback can change when identical feedback is assigned to different conversational roles [13]. This motivates comparing intrinsic and external-role critique while holding the critique content constant.

Compute-aware controls are needed to separate reflection from the benefit of extra calls. One recent study reports that repeated sampling can match or exceed Self-Refine and Reflexion under equal token budgets in its tested settings [44]. Other recent reasoning evidence argues that refinement can outperform resampling under different test-time conditions [8]. The studies use different tasks, models, correction signals, and definitions of budget, so the findings are not directly contradictory. They do show why refinement must be compared with resampling under a stated cost budget.

The number of revisions also matters. Feedback-control analysis describes self-correction as a dynamic process that may improve initially and then cross a stability boundary where further revision becomes harmful [39]. An evaluation should report where the loop stops, what changes between attempts, and how much computation it uses.

2.3 Retrieval-Augmented Generation

Retrieval-augmented generation combines a parametric generator with external non-parametric memory [38]. Retrieved material can provide facts, examples, or domain-specific language that is absent from a prompt. It can also create leakage, copying, and evaluation circularity when retrieved examples overlap with training or assessment material. Its value depends not only on retrieval quality, but also on how the index is built and which data it may contain.

Here, reviews provide examples of register and target level; they are not factual evidence about a plot. This distinction matters because the source corpus contains no review-to-film mapping. Retrieval cannot establish that a generated statement is true of the input film; it can only condition the style and specificity of the response. The index is restricted to R1, excludes R2 and Gold-300, retrieves within the requested level, and retains copy diagnostics. A randomized static few-shot condition separates the effect of receiving examples from the effect of semantic retrieval.

HybridRAG-BN is a close recent Bangla-language adjacency. It addresses knowledge-base question answering (KBQA) by combining BM25 lexical retrieval with BGE-M3 dense retrieval, Gemma generation, and a verifier and refiner obtained by low-rank adaptation (LoRA) fine-tuning of Gemma [2]. Its retrieved knowledge base can provide query-relevant facts, and its verifier participates in answer repair. The index used in this thesis instead provides response exemplars, not plot facts. No language model is fine-tuned, and Verifier-B cannot participate in revision. HybridRAG-BN demonstrates retrieval and verification in Bangla, but it does not address the same data separation or audience-response task.

2.4 Neural and Symbolic Verification

Neural and symbolic evaluators offer complementary properties. Neural classifiers can learn flexible boundaries from contextual representations, but their scores rarely identify an actionable reason for failure. Hand-specified symbolic rules can name observable defects, yet such rules may be incomplete, brittle, or weakly predictive. A neuro-symbolic design must state whether symbolic information changes the decision boundary, explains a neural decision, or guides a subsequent repair; these are different claims and they carry different evidentiary burdens.

SymDiag provides a recent example in which symbolic verification supports explainable diagnosis and repair of multi-step reasoning [15]. Its satisfiability-based setting differs from short Bangla comments, for which no complete formal theory of an acceptable response exists. The useful connection is architectural: outcome assessment can be separated from a diagnosis that names the failed constraint.

Prediction and diagnosis require separate evidence. Symbolic information should affect acceptance only if it improves held-out discrimination beyond the neural score; otherwise, its role is limited to naming observable failures and guiding revision. The experiment tests symbolic gating and symbolic diagnostic feedback separately; the term *neuro-symbolic* is not treated as evidence of predictive benefit.

2.5 Bounded Multi-Agent Workflows

Multi-agent language-model systems divide a task among roles such as planner, retriever, writer, critic, or reviewer. Role separation can make intermediate decisions inspectable and can restrict which component may access a particular tool or dataset. It does not, by itself, guarantee better output. Additional roles can duplicate work, amplify an early error, or consume more tokens without adding independent evidence.

A recent controlled comparison of single-agent and multi-agent RAG systems for repository documentation illustrates this trade-off [54]. The multi-agent workflow achieved strong structural consistency, but a simpler single-agent pipeline obtained comparable lexical quality with substantially lower token cost. Developer-guided planning performed best in that study, suggesting that the value of decomposition depended on where reliable structure entered the workflow rather than on the number of agents alone. Its software-documentation task is not directly comparable with Bangla cinema responses, but the result rules out a generic claim that agentic complexity is inherently beneficial.

The architecture is described as a **bounded multi-agent workflow** with a predefined control flow, not as an autonomous system. Writer and Reflector are model-calling roles; Researcher is a retrieval and tool role that makes no model call; Critic is a deterministic neural-symbolic evaluation role that also makes no model call. Their privileges and transitions are fixed rather than negotiated at run time. The ten-condition experiment tests retrieval, verification, feedback, judging, and resampling mechanisms. It does not include a direct single-agent-versus-multi-agent architecture ablation and cannot establish that a multi-agent architecture is better than a single-agent alternative.

2.6 Verifier Reliability and Proxy Gaming

Learned verifiers can make generation control inexpensive, but they also create a proxy: the system optimizes what the verifier recognizes rather than the intended construct itself. Evaluator Stress Tests formalize this concern through perturbations that separate proxy improvement from performance under an independent criterion [3]. Large language model (LLM) judges introduce related risks, including rubric sensitivity, position effects, and self-preference when evaluator and generator share model families. Fixed responses can receive different measurements after a judge change [65], and rubric-based judges can prefer outputs from their own model family [48]. Multilingual evidence also finds uneven judge reliability across languages [19], while a low-resource Basque study reports weak agreement both with humans and across judges [49]. These findings prevent the hosted same-family Gemma judge from serving as the final Bangla outcome evaluator.

An outcome evaluator must be kept outside the procedure it assesses. Verifier-A is deliberately available during generation, whereas Verifier-B is trained on disjoint data with a different pretrained model and tokenizer and is never allowed to gate, select, critique, or rewrite an output. A widening A–B gap is interpreted as evidence consistent with proxy overoptimization. It is not proof that Verifier-B represents human truth or that human-perceived quality has declined.

2.7 Bangla NLP for This Study

2.7.1 Classification Backbones

Bangla NLP systems vary in pretraining corpus, tokenizer, script coverage, and register. BanglaBERT provides a Bangla-specific pretrained language model and evaluation benchmarks [7], while LaBSE provides language-agnostic sentence embeddings suitable for

multilingual semantic comparison [18]. These resources enable controlled studies in Bangla, but model identity alone does not determine which representation will best reproduce a constructed label.

Recent Bangla classification studies reinforce that caution. MuRIL performs strongly on Bangla emotion detection [45], whereas studies on BanglaBlend and Bangla form classification report advantages for XLM-R or IndicBERTv2 over BanglaBERT in their respective settings [24, 42]. Their tasks and datasets do not settle the verifier choice for this thesis. They show why a Bangla-specific backbone cannot be assumed to be superior from its name alone.

This is especially relevant for short reviews. Sentiment, length, punctuation, and source register may dominate a representation that is later interpreted as engagement or persona. These factors are audited before the construct is named, and the verifier backbones are compared on the task rather than selected from model identity.

2.7.2 Generation and Evaluation Resources

Classification backbones address only half of the low-resource problem. A study that generates Bangla text also depends on what is known about Bangla generation quality and on which instruments exist to evaluate it, and that literature is both thinner and more recent.

Three 2026 resources are directly relevant. A multi-task hallucination evaluation framework for Bengali reports that no prior work had systematically evaluated hallucination in large language models for the language, despite its speaker population [1]. A curated dataset on honorific failures in multilingual Bangla generation targets a failure mode in which output is superficially polite but pragmatically wrong [58]. A benchmark for sociopragmatic and cultural alignment in Bangladeshi social interaction reports that fluency alone does not guarantee socially appropriate language use in a high-context language [59]. Alongside these, work on informal Bangla machine translation documents that informal Bangla is under-resourced relative to formal Bangla [53]. This matters because the review corpus consists of informal comments.

Apparent Bangla fluency cannot replace evaluation by a separately trained verifier and native readers. Register and honorific use also remain possible failure dimensions outside the symbolic checks used in this study. They are reported as limitations of the evaluation.

2.8 Human Evaluation of Controlled Generation

Human evaluation serves two distinct purposes in this thesis: validating the corpus-derived construct and assessing whether generated outputs express the requested level. These pur-

poses require different instruments. Comparative judgments can be more reliable than rating scales for subjective intensity annotation [32], while recent work shows that ratings and comparisons may provide complementary information rather than interchangeable measurements [17]. The elicitation format must therefore match the construct, and the instrument must be tested rather than assumed to be reliable.

The generated-output study uses a forced binary target-level judgment rather than a general quality rating. Agreement is reported through raw three-way agreement and nominal Krippendorff alpha [34], with item-level bootstrap uncertainty. HEDS 3.0 guides transparent reporting of recruitment, evaluated systems, allocation, criteria, ethics, and analysis decisions [5]. Agreement is not treated as validity on its own, and the human study is not used to rank conditions whose per-cell sample is too small.

The composition of the evaluation set also matters. Outcome-blind balanced selection avoids conditioning the human sample on either verifier, a risk noted for metric-guided NLG evaluation sampling [67]. Replicable evaluation depends jointly on item allocation and ratings per item [51], and persistent rater identifiers are needed to represent annotator variation [46]. Work on global and pairwise scoring likewise supports matching the elicitation form to the estimand instead of treating one format as universally best [37]. For three nominal raters, metric-selection guidance further supports reporting Krippendorff alpha with uncertainty and disagreement patterns instead of a universal agreement cutoff [29].

Sections 2.1–2.8 identify the methods used in this thesis and the limits of the evidence behind them. The remaining gap lies in testing those methods under the same data and evaluation constraints. Section 2.9 states that gap directly.

2.9 Research Gap

Table 2.1 summarizes how selected adjacent studies differ from the design used in this thesis.

None of the selected studies combines R1-based retrieval, bounded revision with Verifier-A, an isolated Verifier-B, deterministic diagnosis, and blinded assessment for short Bangla cinema responses. The resulting gap is evaluative rather than algorithmic: whether this combination works when data access, computational cost, negative results, and possible proxy optimization are recorded. This thesis tests that combination without presenting retrieval, verification, symbolic diagnosis, or multi-agent roles as new inventions.

Table 2.1 How the present study differs from adjacent work

Work	Task	Retrieval	Feedback	Symbolic diagnosis	Isolated outcome scorer	Human evidence
Mixture-of-Personas [10]	Population-conditioned text generation	No	Persona conditioning	No	No	Study-specific evaluation
SimAB [50]	Persona-conditioned web A/B prediction	Context documents	Agent interaction	No	No	Historical outcomes and practitioner study
FUDGE [64]	Controlled text generation	No	Learned discriminator during decoding	No	No	Task metrics
Self-Refine [40]	Multi-task refinement	No	Same-model self-feedback	No	No	Task-dependent human and automatic evaluation
Reflexion [57]	Reasoning, coding, and decision tasks	Task-dependent	Verbal feedback or memory	Verbal or heuristic	No	Benchmark outcomes
HybridRAG-BN [2]	Bangla knowledge-base question answering	BM25 and BGE-M3	Fine-tuned verifier and refiner	No	No	Competition token-F1
SymDiag [15]	Multi-step reasoning	No	Symbolic diagnosis supports repair	Yes	No isolated A/B wall	Manually audited diagnosis
Saleh et al. [54]	Repository documentation	Repository retrieval	Reviewer-mediated rewriting	No	No	Manual structural analysis and automatic metrics
Present study	Bangla cinema-response generation	R1 level-specific exemplars	Verifier-A and bounded feedback	Deterministic diagnostic rules	Verifier-B outside the loop	Construct study and blinded output study

2.10 Chapter Summary

Prior work supplies the mechanisms used here, but also shows where they fail: self-correction can be unreliable, retrieval can leak information, and repeated optimization can exploit an evaluator. These risks motivate the data boundaries, matched controls, and separate human and model-based evidence developed in Chapter 3.

Chapter 3

Data and Construct Validation

This chapter explains how unlabelled Bangla reviews were turned into a defensible generation target. Because the corpus contains no engagement-specificity label, the proposed distinction is tested against source, sentiment, length, geometric, and human evidence before it is used in generation. Two data resources serve different purposes. The review corpus supports construct development, retrieval, and verifier training. A separate corpus of Bangla Wikipedia film synopses supplies generation stimuli. The resources are not merged into film–review pairs because the review data contain no movie-title field. Consequently, the study evaluates controlled response generation rather than film-specific audience prediction.

Figure 3.1 summarizes the end-to-end research design and makes the principal isolation constraints explicit. In particular, Gold-300 is evaluation-only, retrieval is restricted to R1, and the sealed Verifier-B reaches generated responses only during independent outcome evaluation.

3.1 Study Design

The study uses a staged quantitative design followed by two human-validation instruments. The Bangla review corpus is first audited and partitioned under fixed data-access rules. Unsupervised representation analysis then proposes a textual distinction, while confound diagnostics and a negative-control corpus region test what that distinction captures. Human judgments finally test whether the retained contrast is perceptible and whether its direction corresponds to engagement specificity. Only after those checks do the two levels become targets for verifier training and controlled generation.

Exploration, model development, and final evaluation use different data. Model and threshold decisions use the permitted R1 development subset and the 30 development plots. The comparative construct study is separate: it uses unseen Region-A reviews outside Gold-

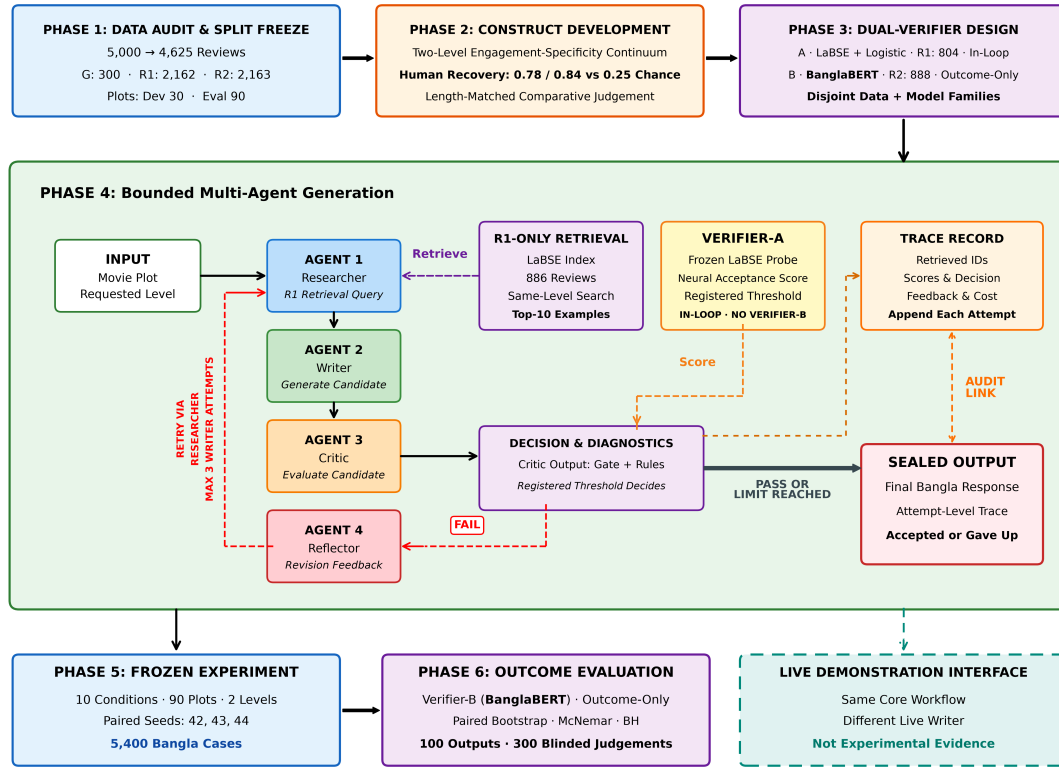


Fig. 3.1 Study design and data-isolation boundaries.

300, including eligible items from R1 and R2. Gold-300 remains confined to the failed ordinal instrument, while the 90 evaluation plots are reserved for the main experiment. Condition comparisons use the same plot, target level, and seed. Verifier-A may influence generation; Verifier-B is used only for final outcome measurement. The global random seed is 42. Additional generation seeds are paired sensitivity blocks rather than independent replications. No language model is fine-tuned. Training is limited to the registered verifier models, logistic-regression, and temperature-scaling artifacts. The resulting evidence concerns controllability and evaluation, not prediction of real audience behaviour.

Numerical results appear here only when they decide whether the construct can proceed. The corpus-provenance analysis determines the admissible data region, and the human studies determine whether the candidate axis can be used. Verifier performance, generation conditions, and final outcome comparisons are reported in their respective later chapters. Keeping them separate prevents evidence for admitting the construct from being mistaken for evidence that the generation framework succeeds.

3.2 Data Sources

3.2.1 Bangla Review Corpus

The primary resource is the *Raw Bangla Movie Review Comment Dataset for Sentiment Analysis and Natural Language Processing*, version 3 [26]. The workbook contains 5,000 rows with a review-text column and a three-class sentiment label. It contains no audience demographics, movie identifier, source URL, or row-level collection provenance. These absences rule out demographic persona modelling and direct comparison between generated responses and real responses to the same film. The raw workbook was treated as read-only. Text processing was intentionally minimal: whitespace was normalized, but Bangla characters were not transliterated, stemmed, or subjected to stopword removal. This preserves the short, informal register required by the contextual encoders and avoids introducing preprocessing artefacts into the construct analysis.

3.2.2 Plot Corpus

Generation stimuli were collected separately from Bangla Wikipedia through the MediaWiki API. The harvest identified 3,135 candidate articles; 124 passed the mechanical quality gate, and 120 were retained after manual review. Each retained row records its source URL, revision identifier, revision timestamp, and CC BY-SA 4.0 metadata; the complete article-revision attribution is provided in the appendix. Plot sections contain between three and twelve sentences, with a median of nine. A seed-42 split assigns 30 plots to development and 90 to the final evaluation surface. Plot text enters neither construct discovery nor verifier training.

3.3 Cleaning and Quality Control

The read-only audit reproduced eight of eleven quantities claimed in the initial pipeline specification. It confirmed the 5,000-row size, sentiment counts, 204 exact duplicate occurrences, 72 reviews shorter than three whitespace tokens, an eight-word median, an 84-word maximum, and the absence of URLs, mentions, and emoji rows. It corrected three claims: there are two null rows, 206 normalized duplicate occurrences, and the rule-based usable count is 4,730 rather than approximately 4,722. The discrepancy arose because the original estimate subtracted null, duplicate, and short-review counts as though the sets were disjoint. Ten short reviews were also duplicates. Cleaning therefore applied the rules sequentially and recorded the actual change at each step. The raw audit counts 204 byte-identical duplicate occurrences;

after whitespace normalization, one additional pair becomes exact, so the sequential cleaning stage removes 205 rows before the normalized-key check removes one further row. Table 3.1 records this sequence.

Table 3.1 Review-corpus audit and cleaning

Stage	Rows after stage	Rows removed	Decision
Raw workbook	5,000	—	Read-only source
Remove missing text or label	4,998	2	One missing text; one missing label
Strip URLs, HTML, and mentions	4,998	0	No matching rows
Normalize whitespace	4,998	0	254 texts changed in whitespace only
Remove exact duplicates	4,793	205	First occurrence retained
Remove normalized duplicates	4,792	1	NFC/whitespace key used only for comparison
Remove reviews shorter than three words	4,730	62	Whitespace-token definition
Near-duplicate control	4,625	105	Applied before the frozen split

Cleaning changed the sentiment distribution from 1,665/1,664/1,670 to 1,513/1,599/1,618 after rule-based cleaning. The three classes did not shrink equally, and no later step assumes equal class sizes. Stable review identifiers were derived from raw row positions before filtering, allowing the frozen map to refer to records without renumbering them. Near duplicates were then controlled using cosine similarity of L2-normalized LaBSE embeddings [18] at the registered threshold of 0.95; 105 additional rows were removed before partitioning.

3.4 Frozen Data Split

Near-duplicate control produced a 4,625-row split surface. The partition was stratified by sentiment and corpus region, rather than by a discovered cluster, because the full-corpus clustering was subsequently shown to recover a source signature. The split map is frozen and cannot be regenerated or reshuffled. Table 3.2 shows the size and permitted use of each partition.

Table 3.2 Frozen review partitions and permitted uses

Partition	Rows	Region A / Region B	Permitted use
Gold-300	300	123 / 177	Human construct evaluation only
R1	2,162	886 / 1,276	Region-A subset supplies the RAG index and Verifier-A pool; non-Gold items were eligible for comparative validation
R2	2,163	888 / 1,275	Region-A subset supplies the Verifier-B pool; non-Gold items were eligible for comparative validation but never retrieval
R1 development subset	200	82 / 118	Threshold and model-development decisions within R1

The development subset is contained within R1; it is not a fourth disjoint partition. Gold-300 is evaluation-only. Retrieval uses R1 exclusively, and more specifically its 886-row Region-A subset; the 1,276 Region-B rows do not enter the core retrieval or verifier path. Verifier-B is developed from the 888-row Region-A subset of R2 and remains unavailable to generation. The later comparative instrument sampled unseen Region-A reviews from the non-Gold pool; it did not place R2 in retrieval or expose Verifier-B to generation. These access rules prevent retrieval, training, and final evaluation data from being mixed.

3.5 Corpus Source Diagnostics

Initial clustering over the full cleaned corpus appeared to recover three groups. A trap-check against sentiment produced an adjusted Rand index (ARI) of 0.1793, suggesting that the partition was not simply the sentiment label. However, inspection of row order and register revealed a sharper explanation: the workbook contains two contiguous corpus regions joined at raw row 1,999.

Region A contains the first 1,999 raw rows and has the varied punctuation and first-person language expected of organically collected comments. Region B contains the remaining 3,001 rows and has a highly uniform sentence-final and lexical signature. All neutral-labelled reviews occur in Region B. Table 3.3 summarizes these differences. The region variable predicts the full-corpus clustering with 93.3% accuracy; its binary ARI is 0.7487 and ϕ is 0.861. The apparent audience grouping was primarily a corpus-source detector.

Because original row-level source metadata were not retained, the precise origin of Region B cannot be reconstructed. The observable source signature is enough to rule out

Table 3.3 Source-region differences in the raw workbook

Diagnostic	Region A	Region B
Rows	1,999	3,001
Danda-terminated reviews	38.7%	99.2%
First-person language	13.5%	0.8%
Exclamation marks	3.4%	0.3%
Median words	9	8
Word types per 1,000 tokens	255.0	127.6
Neutral-labelled reviews	0	1,670

an audience-structure interpretation of the full-corpus partition. Construct development was limited to Region A, while Region B was retained as a negative control.

3.6 Developing the Engagement-Specificity Axis

Reviews were represented with Language-Agnostic BERT Sentence Embedding (LaBSE) [18]. K-means solutions from K=2 to K=8 were compared using prediction strength [60], bootstrap ARI [28], silhouette [52], the gap statistic [61], and cluster-size balance. HDBSCAN provided a density-based check that did not require a fixed K [11, 43]. All clustering was performed in the original embedding space rather than a visualization projection. The prediction-strength decision threshold was fixed at 0.80 before the candidate K table was computed. Tibshirani and Walther describe 0.80–0.90 as a practical threshold range and select the largest K above the chosen threshold [60]. This study pre-specified the lower endpoint of that range; it was not estimated from the observed K table. Because the cited recommendation was demonstrated primarily for well-separated clusters, crossing 0.80 was treated as a reproducibility screen rather than evidence of natural cluster structure.

The diagnostics answer different questions and were not combined into a single post-hoc score. Prediction strength and bootstrap ARI assess reproducibility. Silhouette and the gap statistic assess separation, while HDBSCAN tests whether a density-supported grouping exists. ARI with sentiment tests label rediscovery, and directionless AUC measures whether length or another surface feature can recover the partition. Region B provides a negative control for the possibility that a stable partition can be contentless.

Figure 3.2 displays the four K-dependent diagnostics without collapsing them into a single score. Prediction strength is the registered selection criterion: K=2 is the only candidate above 0.80 in both regions. Bootstrap ARI shows that this cut is reproducible, whereas the very small silhouette values and the absence of a gap-statistic selection show that reproducibility

is not evidence of well-separated natural groups. HDBSCAN is not plotted as a function of K because it does not require a prespecified cluster count; its independent density-based result is reported in Table 3.4.

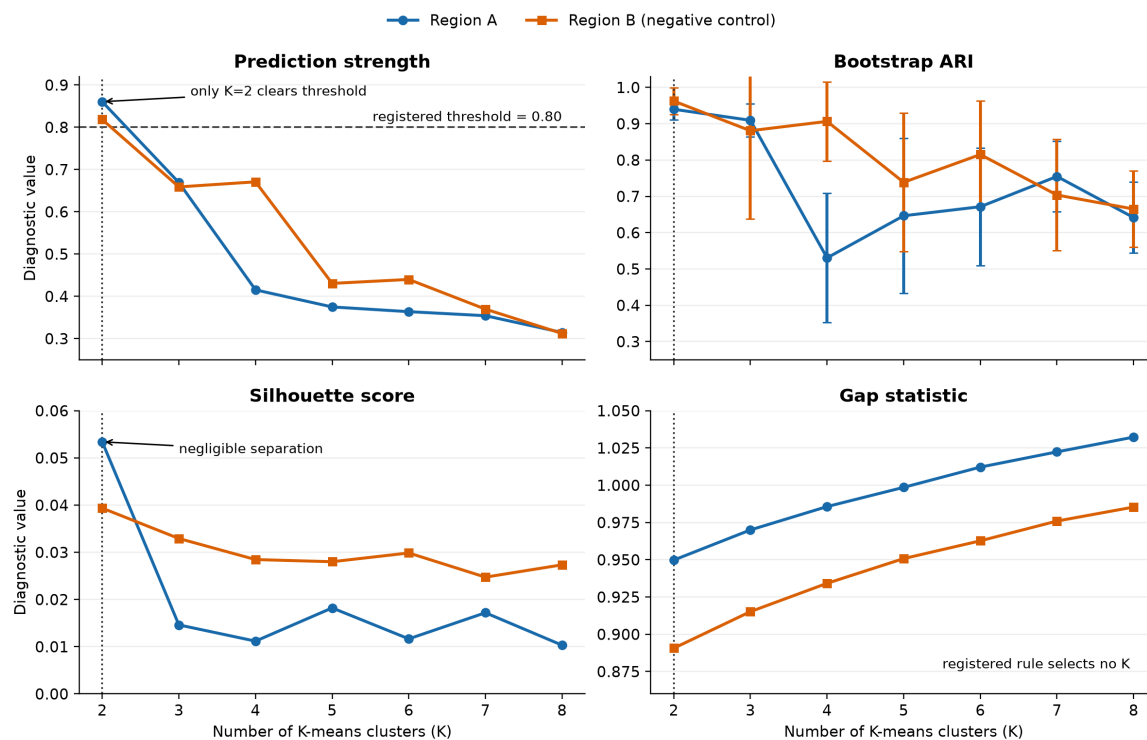


Fig. 3.2 K-selection diagnostics for Region A and the Region-B negative control.

For Region A, $K=2$ was the only solution to clear the registered prediction-strength threshold of 0.80, reaching 0.8605 with bootstrap ARI 0.9399 ± 0.0290 . The observed prediction strength lies between the two endpoints discussed by Tibshirani and Walther: it clears the registered 0.80 rule but would not clear a 0.90 rule. The selection depends on the pre-specified 0.80 cutoff and is not threshold-invariant evidence. The cut was not identical to sentiment (ARI=0.1522). These stability results did not establish discrete structure: silhouette was only 0.0534, the gap statistic selected no K , and HDBSCAN labelled all points as noise. The cut is best understood as a reproducible bisection of a continuum rather than two naturally separated groups. Profiling showed that length was an important but incomplete component of the cut. Word count recovered the assigned level with AUC 0.6764. The longer half averaged 13.12 words and the shorter half 8.85 words; at an equal 4,000-token budget, however, the shorter half contained approximately 1,913 word types against 1,623 in the longer half. A residual classifier improved by 9.81 percentage points over sentiment and

length together, but this estimate is a resubstitution upper bound and lies close to its registered threshold. It is reported as weak descriptive evidence, not confirmatory inference.

Region B demonstrates why stability alone is insufficient. Its $K=2$ cut has prediction strength 0.8183 and bootstrap ARI 0.9624, yet silhouette is 0.0394, HDBSCAN labels 96.7% of observations as noise, and the Region-A length and lexical-richness signatures do not replicate.

Table 3.4 Construct evidence and Region-B negative control

Diagnostic	Region A	Region B	Interpretation
Split-surface rows	1,897	2,728	Separate corpus regions
Selected K	2	2	Both can be bisected
Prediction strength	0.8605	0.8183	Stable in both regions
Bootstrap ARI	0.9399 ± 0.0290	0.9624 ± 0.0362	Stability does not imply content validity
Silhouette	0.0534	0.0394	Negligible separation
Gap statistic	No K selected	No K selected	No positive clusterability evidence
HDBSCAN noise	100%	96.7%	No density-supported groups
ARI with sentiment	0.1522	0.0107	Neither is a sentiment relabelling
Length AUC	0.6764	0.5498	Region-A cut is length-confounded
Richness inversion across length quartiles	4/4	1/4	Content signature does not replicate in Region B

These results reject the interpretation of the two halves as audience personas. They leave open a narrower question: whether people can perceive a consistent textual distinction across the Region-A cut.

3.7 Human Validation

3.7.1 Failed Ordinal-Rating Study

Two human-validation instruments were used, and their outcomes are kept separate. The first asked two native-Bangla annotators to assign a score from 0 to 3 to Gold-300 reviews. Of 600 possible ratings, 598 were returned and 298 items were rated by both annotators. Ordinal Krippendorff's alpha [34] was 0.4970, below the registered reliability floor of 0.667. Although exact agreement was 75.5%, both raters concentrated their judgments on category 2, making the scale insufficiently discriminative. Gwet's AC1 was 0.8705, but it was not used to override the registered alpha gate because AC1 can respond differently under prevalence

imbalance [23, 63]. The result of the first instrument is inconclusive, not evidence that the construct is absent.

3.7.2 Comparative-Judgment Validation

The second instrument used comparative judgments, which can be more reliable than absolute rating scales for subjective intensity [32]. Each of 50 sets contained three reviews from one side of the Region-A cut and one from the other. These items were drawn from Region A while excluding Gold-300, so neither the failed instrument nor its evaluation material was reused. Reviews within a set were matched to a maximum two-word span. Accuracy was evaluated separately for each annotator using exact one-sided binomial tests and Wilson confidence intervals. A length-only heuristic was evaluated on the same sets. A separate direction task asked which member of a cross-level pair referred to something particular in the film.

The length-only heuristic achieved 0.16 against a chance rate of 0.25. The annotators, without being told the name of the construct, identified 39/50 and 42/50 intruders; both answered 34/40 direction items correctly. Table 3.5 reports the results of both instruments. These outcomes determine whether the candidate construct may be operationalized in the subsequent methodology.

Table 3.5 Results of the two construct-validation studies

Instrument	Sample	Result	Interpretation
Ordinal rating	598 ratings (298 items rated twice)	Ordinal $\alpha = 0.4970$; exact agreement = 75.5%	Inconclusive; reliability gate failed
Intrusion, Annotator A	50 length-matched sets	39/50 = 0.780; 95% CI [0.648, 0.872]; $p = 5.7 \times 10^{-15}$	Exceeds 0.25 chance
Intrusion, Annotator B	50 length-matched sets	42/50 = 0.840; 95% CI [0.715, 0.917]; $p = 3.0 \times 10^{-18}$	Exceeds 0.25 chance
Direction, both annotators	40 pairs per annotator	Each: 34/40 = 0.850; 95% CI [0.709, 0.929]; $p = 4.2 \times 10^{-6}$	Specificity direction identified

The two annotators agreed on 70% of intrusion sets and 75% of direction pairs. The study does not treat the pooled 81/100 intrusion responses as 100 independent trials because both annotators judged the same sets. The per-annotator results are the relevant evidence. The panel is small and based on a convenience sample; the implications are discussed with the other methodological limitations later in the thesis.

3.8 Operational Definition

The validated construct is an engagement-specificity continuum represented by the two operational levels in Table 3.6. It is not a demographic persona, natural cluster, or audience segment.

Table 3.6 The two engagement-specificity levels

Level	Definition	Diagnostic question
Level 0	A general or formulaic opinion that may name a broad film element but does not connect the opinion to a particular reason, event, relationship, or narrative detail	Could the response be placed under many different films without changing its meaning?
Level 1	A response that connects an opinion to a particular event, character, relationship, comparison, expectation, or narrative element	Would the response cease to fit if it were moved to a different film?

Sentiment does not define the levels: praise and criticism may occur at either level. Length also does not define them, even though the corpus cut is length-confounded. The operational distinction is comparative and prototypical, reflecting what the human instrument validated. Later chapters use the terms *axis*, *level*, and *cut*, while the variable name `cluster_k2` is retained only where required by frozen computational artifacts.

3.9 Ethics, Reproducibility, and Method Limits

Human annotation involved adult native-Bangla volunteers who participated with informed consent. Responses were stored under coded identifiers, and personal identity and consent records remain outside the public research artifacts. The annotators were university peers known to the researcher, creating a possible social-pressure and convenience-sampling bias. No claim of institutional ethics approval or exemption is made. The two-annotator design also departs from the originally planned three-annotator construct study and is retained as a declared limitation rather than represented as equivalent. Reproducibility is enforced through a read-only raw-data boundary, stable row identifiers, global seed 42, a non-overwritable split map, configuration-driven scripts, and artifact provenance. The Wikipedia plot corpus additionally stores the exact article revisions and licence metadata. Gold-300, R1, R2, and the plot splits retain fixed privileges throughout the subsequent chapters.

The chapter establishes that the Region-A cut is stable and human-recognizable, but several limitations bound that result:

- the source workbook lacks row-level provenance and movie identifiers.
- Region B has a strong non-organic register signature and is excluded from construct development.
- Region A shows negligible cluster separation and 100% HDBSCAN noise.
- the retained cut remains associated with sentiment and length.
- the successful human instrument used only two annotators from a convenience sample.
- human recognizability of a textual distinction does not establish real-world audience segments or film-specific response validity.

In generated text, requested level also remains recoverable from length alone at AUC 0.91–0.99. Consequently, subsequent chapters may claim control of a human-recognizable response style, but not length-neutral control, audience simulation, or prediction of authentic audience reception.

3.10 Chapter Summary

The audit reduced 5,000 raw reviews to a frozen 4,625-row analysis surface and exposed a major source confound. Region A supported a reproducible two-way cut, but not discrete natural groups. Two native-Bangla annotators recognized the contrast in length-matched comparisons. The defensible construct is therefore a two-level engagement-specificity axis, which the next chapter operationalizes with separate in-loop and outcome verifiers.

Chapter 4

Verifier Development and Isolation

The system uses two verifiers so that the model guiding revision does not also determine the final result. Verifier-A guides generation; Verifier-B scores only completed outputs. Both reproduce the operational levels developed in Chapter 3; human recognition is assessed separately.

4.1 Verifier Design and Data Separation

Verifier-A is the in-loop verifier. It scores drafts, supports acceptance decisions, and helps select among generated alternatives. Verifier-B is used only for final evaluation: it is excluded from retrieval, prompting, feedback, candidate selection, and regeneration. The final analysis tests whether gains against the in-loop scorer also appear under a model that did not influence generation. Cross-family evaluation has similarly been used to detect divergence between an optimised proxy and independent reference models [41].

The two verifiers also differ in their training data and model families. Verifier-A is trained on the labelled R1 partition, while Verifier-B is trained on the disjoint R2 partition. Both are evaluated on the same 82-item development slice so that their predictions can be compared on identical texts. Gold-300 is excluded from training, calibration, retrieval, prompting, and threshold selection. Table 4.1 summarizes these data privileges.

The development slice is held out from both verifier-training sets, but it serves several purposes: learning-rate selection in the ablation, temperature fitting, and the evaluation reported in this chapter. The reported scores are development-set estimates, not independent final-test performance.

Table 4.1 Data available to each verifier

Component	Training partition	Level 0/1	Evaluation partition	Role
Backbone ablation	R1 training subset, $n = 804$	481/323	dev-82 (53/29)	Compare the seven evaluated model configurations
Verifier-A	R1 training subset, $n = 804$	481/323	dev-82 (53/29)	In-loop scoring and selection
Verifier-B	R2 Region-A subset, $n = 888$	531/357	dev-82 (53/29)	Outcome-only evaluation

Note. Dev-82 is the labelled Region-A portion of the 200-item R1 development subset and is excluded from the 804 Verifier-A training items. R2 is disjoint from R1. Gold-300 is not used in any experiment reported in this chapter.

4.2 Backbone Results and Construction Circularity

Seven model configurations were evaluated because earlier Bangla classification studies do not identify a consistent leading backbone. MuRIL, XLM-R, and IndicBERTv2 have each been reported as the strongest model in related Bangla tasks [45, 24, 42]. Evidence that fine-tuned small models can outperform zero-shot generative classifiers also supports using a trained discriminative verifier for this task [9]. The comparison included Bangla-specific, Indic, multilingual, contrastive, and entailment-based alternatives. The five conventional fine-tuning models used four epochs, batch size 16, a maximum sequence length of 128, two learning rates (2×10^{-5} and 3×10^{-5}), and seeds 42–46. SetFit–LaBSE and BERT-NLI followed their respective training procedures [36, 62].

Macro-F1 was used because the two levels are imbalanced. Variation across seeds is reported as sensitivity rather than as an inferential decision rule [21, 12, 20]. The registered comparison used paired bootstrap tests with 10,000 resamples and Benjamini–Hochberg correction across all 21 model pairs [16, 6]. Table 4.2 reports the model results together with the diagnostic baselines used to interpret them.

None of the 21 pairwise comparisons remained significant after correction; the smallest unadjusted p -value was 0.0960. A robustness analysis pooling both learning rates produced the same result. BanglaBERT was retained using the predetermined non-performance tie-break, not because the experiment established its superiority.

4.2.1 Construction Circularity

The frozen LaBSE probe changed how the backbone results had to be read. Chapter 3 formed the two operational levels by applying K-means to LaBSE embeddings. A linear

Table 4.2 Model and baseline performance on dev-82

Model or baseline	Mean macro-F1	Seed SD	Selected learning rate
<i>Evaluated models</i>			
BanglaBERT	0.9647	0.0209	3×10^{-5}
SetFit–LaBSE	0.9590	Not estimable	Not applicable
IndicBERTv2	0.9560	0.0156	3×10^{-5}
MuRIL	0.9421	0.0391	3×10^{-5}
mBERT	0.9402	0.0125	2×10^{-5}
XLM-R	0.9360	0.0219	3×10^{-5}
BERT-NLI	0.9298	0.0165	3×10^{-5}
<i>Diagnostic baselines</i>			
Majority classifier	0.3926	—	—
Length rule	0.6197	—	—
Frozen LaBSE with logistic regression	0.9866	—	Fixed defaults

classifier trained on those same embeddings reached 0.9866 macro-F1, exceeding the highest fine-tuned mean of 0.9647. Figure 4.1 places this baseline beside the seven evaluated models.

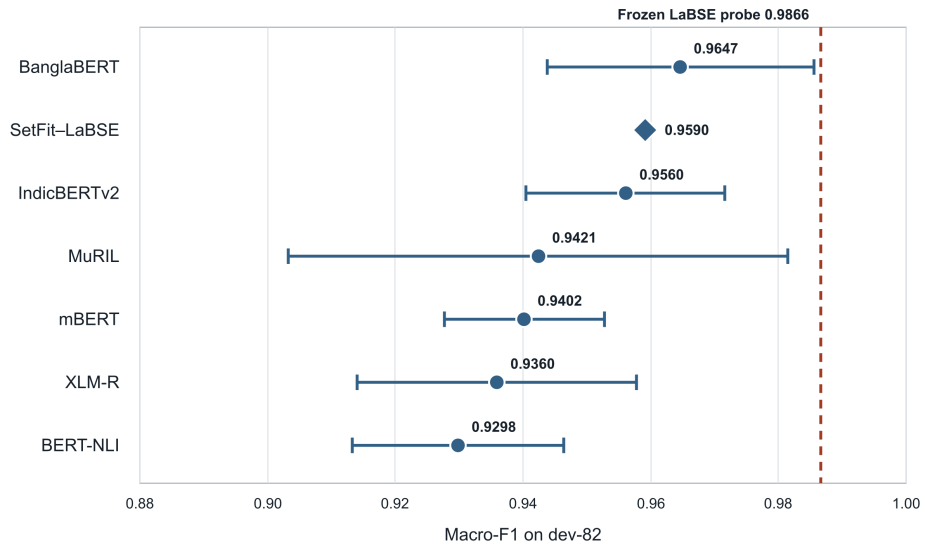


Fig. 4.1 Backbone performance and the frozen-LaBSE baseline on dev-82.

The label is almost linearly recoverable from the representation used to construct it. For that reason, the ablation cannot support a claim about general backbone quality. Its near-ceiling scores show label reproduction, not independent validation of the construct, and the tie should be understood in that context. The length only rule reached 0.6197, compared with 0.3926 for the majority classifier. Length is consequently part of the operational distinction, although the residual and human-validation analyses in Chapter 3 show that it does not

explain the distinction completely. In particular, annotators recovered the intended direction in length-matched comparisons without access to either verifier. This circularity matters when Verifier-A is placed inside the generation loop. A generator may raise its score by moving text toward the preferred region of LaBSE space without producing an improvement that transfers to another evaluator. Verifier-B uses a different model family and different training data, and it is applied only after generation.

4.3 Final Verifier Configurations

4.3.1 Verifier-A

Verifier-A combines the frozen `sentence-transformers/LaBSE` encoder [18] with an L2-regularised logistic-regression head trained on 804 R1 items. The embeddings are L2-normalised, and the classifier uses fixed library settings rather than development-score-based hyperparameter selection. On dev-82, it reproduced the circularity baseline exactly: macro-F1 was 0.9866, with one Level-1 item classified as Level 0. The model is inexpensive enough for repeated calls during generation, but its high score should be interpreted narrowly. It shows that the operational boundary can be recovered from its originating embedding space. It does not establish human-ground-truth validity, and the same geometric relationship that makes the model accurate may also make it vulnerable to optimisation against its score.

4.3.2 Verifier-B

Verifier-B is a `BanglaBERT` model [7] trained from scratch on the 888 labelled R2 items using the same four-epoch setup evaluated in the backbone comparison. It uses a fixed learning rate of 2×10^{-5} . The learning rate was not selected on dev-82, avoiding another validation-based choice on an already small and repeatedly used slice; recent evidence also shows that validation-optimal hyperparameters do not always transfer better than defaults in small holdout settings [56]. The final model uses seed 42, which was fixed before training rather than chosen from the best of five runs.

The stored model reached 0.9597 macro-F1 on dev-82, making three errors. Across seeds 42–46, the mean was 0.9674 ± 0.0158 , with a range from 0.9448 to 0.9866. This range is a sensitivity result, not evidence for comparing Verifier-B with Verifier-A. The models use different training partitions and architectures, and dev-82 is too small and too heavily reused to support a ranking. Table 4.3 summarizes their assigned roles, training data, model designs, and development results.

Table 4.3 Roles and data boundaries of the final verifiers

Dimension	Verifier-A	Verifier-B
Function	In-loop scoring and candidate selection	Outcome-only evaluation
Training data	R1, $n = 804$ (481/323)	R2, $n = 888$ (531/357)
Model	Frozen LaBSE + logistic regression	Fine-tuned BanglaBERT
Development macro-F1	0.9866 (1 error)	0.9597 (3 errors; seed 42)
Seed sensitivity	Deterministic fit	0.9674 ± 0.0158 across five seeds
Access during generation	Permitted	Prohibited

Figure 4.2 shows the error distribution on the shared development items. Verifier-A made one false negative. Verifier-B made one false positive and two false negatives. The reference labels are the operational labels from Chapter 3 rather than independent human-gold judgements.

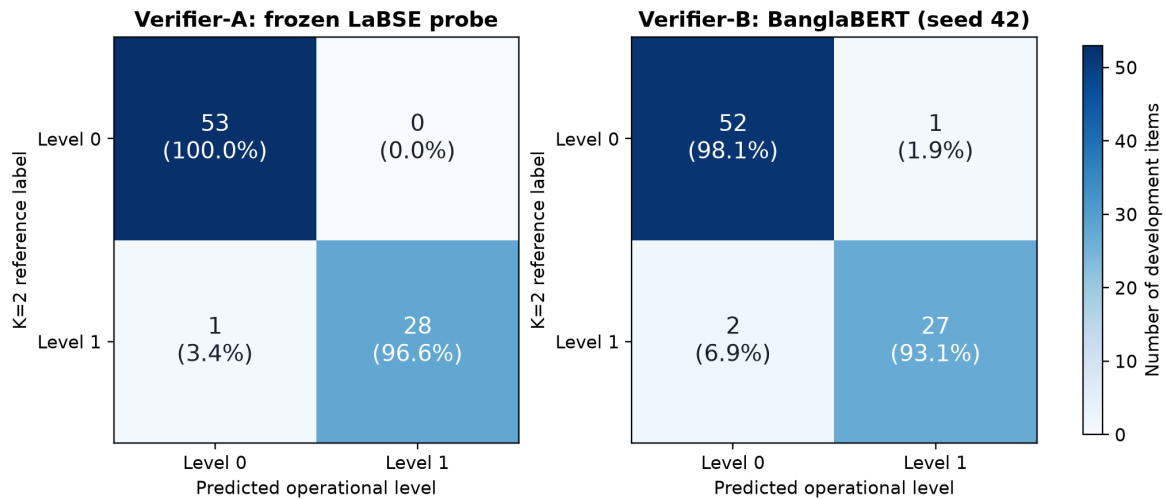


Fig. 4.2 Confusion matrices on dev-82.

4.4 Probability Calibration

Temperature scaling was fitted separately for each verifier [22]. Logistic regression was not assumed to be calibrated merely because it performed well as a classifier; frozen-encoder studies report that a head can rank highly in accuracy while remaining weaker on calibration measures [66]. A single-parameter method was preferred because more flexible calibrators can become unreliable when the calibration sample is small [4]. Five bins were used for expected calibration error (ECE), and bootstrap intervals were calculated for the reduction

in ECE. Since temperature fitting and assessment use the same 82 items, the results remain descriptive. Table 4.4 reports the fitted temperatures and the before-and-after calibration measures.

Table 4.4 Calibration results on dev-82

Verifier	Temperature	Metric	Before	After	Interpretation
Verifier-A	0.1092	ECE	0.1184	0.0054	Reduction 0.1130; 95% CI [0.0743, 0.1349]
		Brier score	0.0306	0.0093	Descriptive reduction
		Negative log-likelihood	0.1515	0.0282	Descriptive reduction
Verifier-B	1.0995	ECE	0.0164	0.0100	Reduction 0.0065; 95% CI [−0.0066, 0.0070]
		Brier score	0.0278	0.0273	Descriptive reduction
		Negative log-likelihood	0.1101	0.1088	Descriptive reduction

Verifier-A’s temperature is below one, sharpening probabilities that were underconfident on this slice. Its ECE reduction is supported by the bootstrap interval. Verifier-B required only mild softening, and its interval includes zero; calibration improvement could not be established for that model. Later chapters treat Verifier-B scores as fixed model outputs rather than as probabilities with demonstrated out-of-sample calibration.

4.5 Isolation, Reproducibility, and Limits

The separation between the verifiers is enforced at several points. R1 and R2 identifiers are checked for overlap, both training sets are checked against dev-82 and Gold-300, and Verifier-B’s configuration records its outcome-only role. Static checks prevent generation modules from importing Verifier-B, while generation records are rejected if they contain a Verifier-B score. These controls preserve the central comparison: both models evaluate the same generated texts, but only Verifier-A can affect how those texts are produced. The shared development set does not weaken the training-data separation. None of its 82 items appears in either verifier’s training data. Its use for both models is necessary for a like-for-like descriptive comparison, although its repeated use limits generalisation claims.

Dev-82 is held out from training but reused for model selection, calibration, and reporting, so it is not an independent final test set. The operational labels come from LaBSE geometry rather than human-gold verifier labels. The SetFit implementation did not produce effective seed or learning-rate variation, and calibration was fitted and assessed on only 82 items.

Calibration improved for Verifier-A on this slice, but the corresponding improvement was not established for Verifier-B. The chapter therefore makes no claim about general backbone superiority, human-ground-truth accuracy, or out-of-sample calibration.

4.6 Chapter Summary

Verifier-A reproduces the LaBSE-derived boundary from R1 and guides generation; Verifier-B learns from disjoint R2 data and scores only sealed outputs. The ablation did not identify a superior backbone, while the near-perfect frozen LaBSE probe exposed construction circularity. Chapter 5 therefore uses Verifier-A for control and preserves Verifier-B for outcome evaluation.

Chapter 5

Bounded Neuro-Symbolic Workflow

This chapter turns the engagement-specificity levels and verifier roles into a bounded generation workflow. The Researcher retrieves examples, the Writer drafts a response, the Critic evaluates it, and the Reflector prepares feedback. Transitions, retries, data access, and scoring privileges are fixed before generation.

5.1 Roles and State

5.1.1 Functional Roles

The term *multi-agent* refers here to functional decomposition and message passing, not to four autonomous or independently trained models. Within the proposed four-role workflow, only the Writer and Reflector make generative-model calls. The Researcher performs deterministic retrieval, while the Critic applies fixed local scoring functions. Table 5.1 summarizes the information and privileges assigned to each role.

Table 5.1 Roles, inputs, outputs, and model access

Role	Function	Inputs	Output	Model call
Researcher	Retrieve ten same-level R1 examples	Plot, target level and optional failed-feature terms	Ranked exemplars	No
Writer	Produce or revise the Bangla response	Plot, level definition, exemplars and available feedback	Candidate response	Yes
Critic	Apply the gate and symbolic diagnostics	Candidate response and target level	Gate result and named failures	No
Reflector	Prepare a bounded correction brief	Draft, target level and permitted diagnostic information	Feedback for the next attempt	Yes

Verifier-A is available to the Critic because it controls acceptance in the neural-loop conditions. Verifier-B is absent from every role and is applied only after an output has been sealed. This separation makes it possible to test later whether improvement against the in-loop verifier transfers to an evaluator that did not influence generation. Figure 5.1 shows the fixed routing. A passing response is emitted immediately. A failed response may be revised until the three-attempt limit is reached. If no attempt passes, the response with the highest gate score is emitted and the trace records that the attempt budget was exhausted.

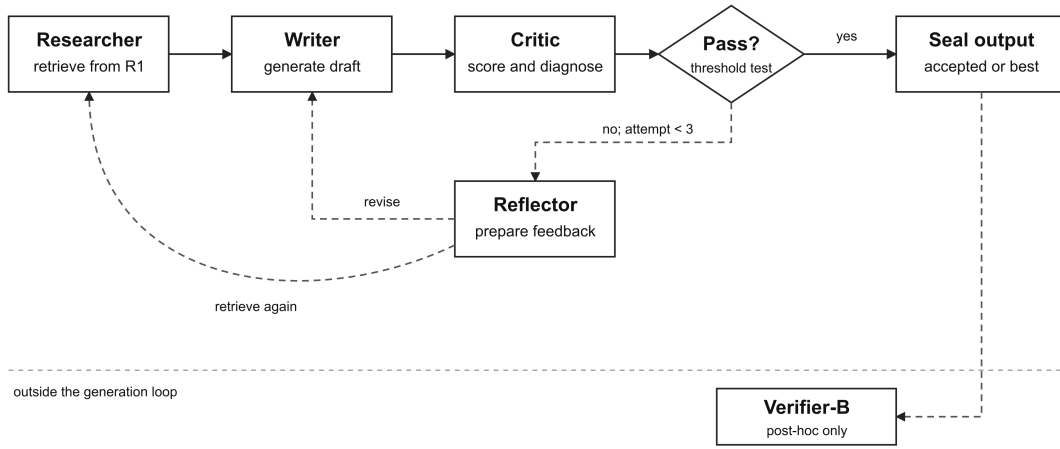


Fig. 5.1 Bounded generation workflow and post-hoc Verifier-B evaluation.

5.1.2 Shared State and Trace

For plot p , requested level l , and attempt t , Equation 5.1 records the controller state as

$$s_t = (p, l, E_t, y_t, a_t, r_t, f_t, c_t), \quad (5.1)$$

where E_t is the retrieved evidence, y_t the current draft, a_t the Verifier-A score, r_t the symbolic diagnostics, f_t the feedback available to the next attempt, and c_t the accumulated logical cost. Verifier-B is not part of this state because no generation decision may consult it. Each attempt is written as a new trace record before the controller advances. The record contains the retrieved identifiers, generated response, scores, gate decision and feedback associated with that attempt. The previous draft and feedback are passed explicitly to the next **Writer** call, which distinguishes revision from independent resampling.

The controller can retrieve, generate, evaluate, request feedback, retry, accept, or emit the best available response when the attempt limit is reached. It cannot change the threshold,

increase the retry limit, alter its data access, select new tools, or change the experimental condition. The framework is therefore an evaluator–optimizer workflow with predefined control flow rather than an autonomous agent that plans its own trajectory.

5.2 Generation and Retrieval

5.2.1 Bounded Generation Loop

Algorithm 1 gives the common loop used by the retrieval-based verifier conditions. The gate may use the neural or symbolic score depending on the experimental condition. Named symbolic failures are supplied to the Reflector only when symbolic feedback is enabled.

Algorithm 1: Bounded verifier-guided generation

Input: plot p , target level l , gate threshold τ , gate type g , symbolic-feedback flag h

Output: emitted response, attempt trace, exhaustion flag

Constants: maximum attempts $T = 3$; retrieved examples $k = 10$;

$t \leftarrow 1$; feedback $\leftarrow \emptyset$; candidates $\leftarrow \emptyset$;

while $t \leq T$ **do**

 Set the plot as the retrieval query anchor;

if $t > 1$ *and* h *is enabled* **then**

 └ augment the query with the previous failed-feature terms;

 Retrieve k same-level R1 examples from the requested level;

 Generate a response from the plot, level definition, examples and feedback;

 Compute the Verifier-A score and symbolic diagnostics;

 Select the gate value specified by g ; append the candidate and attempt trace;

if gate value $\geq \tau$ **then**

 └ emit the current response and stop;

if $t = T$ **then**

 └ emit the candidate with the highest gate value; record exhaustion; stop;

 Produce bounded feedback and increment t ;

Writer generation and non-terminal reflection are the only generative-model calls. A response accepted on its first attempt costs one logical call. A second attempt costs three calls: two Writer calls and one Reflector call. Reaching the third attempt costs five calls. No Reflector call is made after the final attempt because no later Writer call could use its feedback. If several attempts share the highest gate value, the earliest is emitted.

5.2.2 Retrieval and Prompt Construction

The retrieval index contains 886 Region-A reviews from R1: 534 at Level 0 and 352 at Level 1. The reviews are represented with L2-normalized LaBSE embeddings, allowing inner-product search to operate as cosine similarity [38]. The index contains no R2 or Gold-300 identifiers. Its membership is checked against the frozen split before the vectors are stored. Each request retrieves the ten nearest examples from the requested level. The level restriction is applied within the query rather than after retrieval, ensuring that every retrieval-conditioned prompt receives ten examples. The plot remains the query anchor on every attempt. When symbolic feedback is enabled, failed-feature terms augment the plot query but never replace it.

All experimental conditions use the same prompt renderer. Zero-shot generation omits examples and feedback; retrieval conditions add examples; revision conditions add the feedback permitted by their treatment. This shared renderer prevents differences in task wording from being mistaken for effects of retrieval or verification. Bangla text is preserved without transliteration or stemming, and no language model is fine-tuned. Both requested levels use a 20-word output ceiling. The common ceiling reduces the length difference observed during construct development but does not remove it: on the length-controlled development generations, word count alone recovers the requested level with AUC 0.9111. Length diagnostics therefore accompany the later results, and the framework does not claim length-neutral control [31].

5.3 Neural Gating and Symbolic Diagnosis

5.3.1 Separation of Decision and Feedback

Verifier-A supplies the primary acceptance score. The symbolic component evaluates eleven interpretable features and identifies which checks fail. This external diagnostic signal is used because self-correction without external feedback can leave errors unchanged or make a response worse [27, 30]. The symbolic features help specify what should be reconsidered; they are not treated as ground truth.

Whether the symbolic score should also contribute to acceptance was evaluated on the development generations. A combined score $g = wa + (1 - w)r$ was tested over 21 values of w , where a is the Verifier-A score and r the symbolic score. Every plot-grouped validation fold selected $w = 1$, corresponding to neural-only gating. At the same time, changing w altered the pass/fail decision for many responses. Table 5.2 shows that the symbolic component affected decisions without improving held-out discrimination.

Table 5.2 Neural and symbolic signals on development outputs

Condition	Symbolic AUC	Neural AUC	Length AUC	Decisions changed	Held-out result
Length-controlled	0.3417	0.8333	0.9111	50.8%	$w = 1$ in all folds; mean ΔAUC $= 0.0000$
Free-length	0.0656	0.8658	0.9894	39.2%	$w = 1$ in all folds; mean ΔAUC $= 0.0000$

The symbolic scorer reached macro-F1 0.5150 under stratified five-fold cross-validation, compared with 0.6570 when assessed on the same 82 rows used for fitting. This gap, together with Table 5.2, does not support symbolic adjudication. The primary loop therefore uses Verifier-A for acceptance and retains symbolic features for diagnostic feedback. A separate symbolic-loop condition remains as a control, using a threshold of 0.1816651 chosen to reproduce the neural gate’s 39/60 first-attempt acceptance count on the development cases.

5.3.2 Acceptance Threshold and Retry Limit

The Verifier-A threshold was selected on 60 development plot-level cases. Each case had a forced three-attempt trace, allowing every possible stopping threshold to be reconstructed without generating new responses. Candidate thresholds were the observed Verifier-A scores rather than an arbitrary uniform grid. The selection objective in Equation 5.2 follows calibrated cost-aware routing by balancing outcome quality and logical cost [33]:

$$\tau^* = \arg \max_{\tau} \frac{Q_B(\tau) - \alpha_{\text{lo}}}{\mathbb{E}[\text{calls} \mid \tau]}, \quad (5.2)$$

where $Q_B(\tau)$ is measured by Verifier-B after the forced traces were generated. The one-call retrieval endpoint defines $\alpha_{\text{lo}} = 0.640501$, while forced generation of all three attempts defines the higher-cost reference $\alpha_{\text{hi}} = 0.866272$. Verifier-B evaluates the reconstructed policies but never supplies a score during generation. Table 5.3 compares the selected threshold with the two measured endpoints.

The selected threshold captures 71.63% of the Verifier-B gain between the one-call and forced-three endpoints while using an average of two logical calls. Under this policy, 39 cases stop after the first attempt, 12 after the second and one after the third; eight reach the ceiling without passing. The same threshold is used for both requested levels. A permutation test did not detect a difference in attempt-one gate scores between the two levels ($p = 0.4687$), but this does not establish equal difficulty.

Table 5.3 Development operating points for threshold selection

Quantity	One-call retrieval	Selected policy $\tau^* = 0.4384071$	Forced three attempts
Verifier-B outcome score	0.640501	0.802219	0.866272
Mean logical calls	1.000	2.000	5.000
First-attempt acceptance	1.000	0.650	Not applicable
Final acceptance	1.000	0.867	Not applicable
Exhausted cases	0	8/60	Not applicable
Mean emitted-attempt index	1.000	1.367	1.867

Figure 5.2 shows the quality–cost frontier and the forced-attempt score pattern. The first revision improves the mean Verifier-A and Verifier-B scores by 0.0872 and 0.0890. The second revision reduces them by 0.0627 and 0.0180, respectively. These paired development traces support a bounded retry policy but cannot show whether a fourth attempt would help.

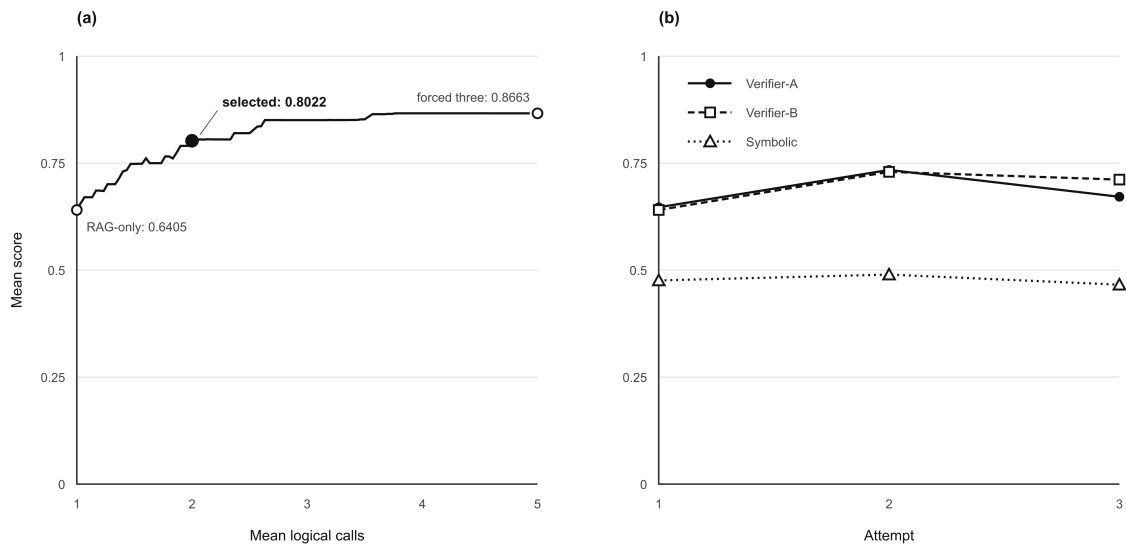


Fig. 5.2 Development quality–cost frontier and forced-attempt score trajectories.

The eight exhausted development cases form a complete census at the selected threshold. One response was off-topic, two required an additional category not included in the predefined taxonomy, and five showed no observable error covered by that taxonomy. The cases were coded once and reviewed by the researcher; no independent second coding was performed. This small, single-coded census is descriptive and does not estimate failure prevalence.

5.4 Experimental Conditions

The main experiment contains ten conditions produced through the same prompt renderer and data-access rules. Table 5.4 specifies the evidence, decision signal, revision mechanism and call ceiling for each condition. Verifier-B is unavailable throughout generation.

Table 5.4 The ten experimental conditions

Condition	Evidence	Decision signal	Revision or selection	Max. calls
Zero-shot	None	None	First draft emitted	1
Static few-shot	Ten deterministically randomized same-level R1 examples	None	First draft emitted	1
RAG-only	Top-10 same-level R1 retrieval	None	First draft emitted	1
RAG + neural loop	Top-10 same-level R1 retrieval	Verifier-A, $\tau = 0.4384071$	Generic feedback and re-retrieval	5
RAG + symbolic loop	Top-10 same-level R1 retrieval	Symbolic score, $\tau = 0.1816651$	Named failures and query augmentation	5
Neural gate + symbolic feedback	Top-10 same-level R1 retrieval	Verifier-A, $\tau = 0.4384071$	Neural result, named failures and query augmentation	5
Intrinsic self-critique	Top-10 same-level R1 retrieval	Model-generated critique	Critique in assistant role	3
External-role self-critique	Top-10 same-level R1 retrieval	Model-generated critique	Same critique in user role	3
Hosted-judge loop	Top-10 same-level R1 retrieval	Judge PASS/FAIL and target-fit score	Bounded judge feedback	6
Blind resampling	Top-10 same-level R1 retrieval	Verifier-A within matched token budget	Independent candidates; no revision	Up to 5

The two self-critique conditions use the same critique text and differ only in message role. A digest check confirms that the critique bytes match before the pair is accepted. This contrast tests whether placement of feedback in the conversation changes the apparent self-correction effect [40, 57]. The hosted judge is `gemma-4-26b-a4b-it`, accessed through the provider’s interaction interface with seed 42, high reasoning effort and a 512-token output limit. It sees the plot, requested level, draft and evaluation rubric, but no Verifier-A or Verifier-B score. It is a hosted same-family control, not an independent outcome evaluator. The eight retrieval-based conditions from RAG-only through blind resampling share the same initial retrieval-conditioned response for each plot–level–replicate key. This pairing removes irrelevant first-draft variation while each condition retains its own subsequent calls and cost.

5.4.1 Matched-Budget Blind Resampling

Blind resampling separates the benefit of revision from the benefit of spending more generation budget. Five independent retrieval-conditioned responses are generated in a fixed order. For each case, only the largest prefix whose cumulative model-token count fits within

the realized token cost of the neural-gate/symbolic-feedback loop is eligible for selection. Verifier-A then selects the highest-scoring response within that prefix. The eligible set is always a prefix in generation order; candidates cannot be chosen after their scores are observed. The budget includes prompt and completion tokens from Writer and Reflector calls under the same generator. It is a model-token comparison rather than a claim of equal hardware FLOPs, latency, or functional computation. If the loop budget cannot fund even one candidate, the matching procedure raises an error instead of silently falling back to an unmatched result. This control follows verifier-ranked sampling [14] while addressing evidence that refinement and resampling advantages can change when compute is matched [44, 8].

5.5 Execution, Reproducibility, and Limits

The Writer is `google/gemma-3-12b-it`, loaded locally with 4-bit NormalFloat quantization. Generation uses a maximum of 80 new tokens, temperature 0.8 and top-p 0.9 under the common 20-word instruction. Every evaluation plot, requested level and condition is crossed with seeds 42, 43 and 44. The resulting design contains $90 \times 2 \times 10 \times 3 = 5,400$ Bangla cases. The seeds are paired sensitivity blocks rather than independent experimental replications.

Before generation begins, the runner checks the 90-plot set, R1-only retrieval manifest, condition list, expected case keys and Verifier-B exclusion. Each completed case records its plot identifier, requested level, condition, seed, attempt count, emitted response, logical calls, token use, stopping status and attempt trace. Checkpoints are append-only, and each case key may occur only once. Verifier-B scoring runs later as a separate process and joins to the sealed outputs by case key.

The live demonstration interface is separate from the frozen experiment. It uses hosted services and an additional operational support check, does not write user plots to the repository, and contributes no evidence to the thesis results.

The threshold and symbolic-weight analyses use development generations rather than the final evaluation surface. The forced traces contain three drafts for every development case; their attempt-to-attempt changes do not describe only the cases that would continue under the selected policy, and they cannot establish an optimal retry ceiling beyond three attempts.

The review corpus has no movie identifier, so retrieval supplies level-matched language examples rather than audience evidence about the film named in the prompt. Requested level remains strongly recoverable from response length despite the common word ceiling. The symbolic scorer is trained on only 82 labelled rows and does not improve held-out discrimination when combined with Verifier-A. Its role is therefore limited to diagnosis.

The failure census contains only eight single-coded cases, and the blind-resampling control matches realized model tokens rather than hardware computation.

5.6 Chapter Summary

The bounded workflow retrieves only from R1, uses Verifier-A for acceptance, and keeps Verifier-B outside generation. Symbolic checks name problems but do not control the primary gate because they added no held-out discrimination. The selected threshold balances development-set outcome quality against logical calls. Chapter 6 compares this workflow with the nine registered alternatives on the frozen 5,400-case surface.

Chapter 6

Results and Analysis

This chapter reports the Bangla generation results, including target-level performance, cost, verifier divergence, human judgments and error cases.

6.1 Design and Statistical Tests

The experiment tests whether short Bangla cinema responses can be generated at a requested engagement-specificity level. Level 0 is a general or formulaic reaction; Level 1 refers to a specific event or construction element. This is a control task, not audience prediction. The corpus has no film identifier, so the outputs cannot be matched to reactions written about the same film.

The evaluation set contains 90 held-out plot synopses, two requested levels, ten conditions and three generation seeds (42, 43 and 44), giving 5,400 outputs. Each condition contributes 540 outputs, divided equally between the two levels. The seeds serve as paired blocks rather than independent experiments, and no best seed is selected. Chapter 5 describes the generation settings and the ten conditions. Table 5.4 defines the ten conditions. Retrieval uses only R1. Verifier-A may guide the registered loops and blind-resampling selector, whereas Verifier-B is sealed from generation and used only for outcome scoring. This separation is needed for the divergence analysis in Section 6.3.

The primary outcome is Verifier-B target probability $p_B(l \mid y)$; binary target match uses its fixed 0.5 decision point. Cost, exhaustion, diversity, length, feature-space similarity and blinded human judgments are reported separately. The primary statistical family contains exactly nine paired comparisons, each active condition against zero-shot. Cases are paired by plot, requested level and generation seed. Uncertainty for the continuous outcome is estimated with 10,000 paired bootstrap resamples, following paired significance-testing guidance for NLP [16]. Binary target match is checked with McNemar’s test, and the nine-

comparison family is corrected with the Benjamini–Hochberg procedure [6]. The design supports comparisons with zero-shot, not a ranking among active conditions. The three seeds are paired blocks, not candidates for selection.

The neural-plus-symbolic versus neural-only contrast in Section 6.2.2 and the same-item human comparison in Section 6.4 are post-hoc and exploratory. Verifier-B scores on all 5,400 outputs remain the primary analysis.

6.1.1 Run Completeness

The final dataset contains exactly 5,400 unique case keys. There are no missing, extra, or duplicate cases, and all 5,400 outputs received a separate Verifier-B score. The run used 7,068 local generation calls and 654 hosted judge calls. Verifier-B is absent from every generation record. These checks establish completeness and isolation, not model quality. Every analysis in this chapter uses this fixed dataset. The exploratory comparisons and qualitative examples neither rerun generation nor rescore the responses.

6.2 Results Across Conditions

Table 6.1 reports all 20 condition-level cells, with 270 outputs per row. Verifier-B supplies both outcome columns. Calls and tokens cover the same-model work charged to each condition; hosted-judge calls are separate, and blind resampling is reported at its matched token budget.

Every condition has higher automatic accuracy at Level 1 than at Level 0. Zero-shot shows the widest gap (0.8074 versus 0.3296); neural-plus-symbolic feedback narrows it (0.9593 versus 0.7333) but does not remove it. The symbolic-only loop matches RAG-only at Level 0, is slightly lower at Level 1, and exhausts its budget on 53.33% of Level-1 cases. The two self-critique controls always use three calls, while the neural-gated loops stop earlier on average. These are descriptive patterns; the planned tests compare each condition only with zero-shot.

6.2.1 Differences Between Levels

Length may contribute to the Level-0 deficit. In region A, Level 0 is longer than Level 1 (13.12 versus 8.85 words), and word count alone reaches 0.6197 macro-F1. The development generations reverse that relation (11.47 versus 16.23 words), with requested level recoverable from length at AUC 0.9111. Length and specificity are therefore entangled in both the construct and its measurement [31].

Table 6.1 Verifier-B outcomes and generation cost by condition

Condition	Level	Mean p_B	Target accuracy	Calls	Tokens	Exhausted
Zero-shot	0	0.3513	0.3296	1.000	590.3	0.0000
Zero-shot	1	0.7794	0.8074	1.000	601.4	0.0000
Static few-shot	0	0.6109	0.6037	1.000	823.0	0.0000
Static few-shot	1	0.8100	0.8407	1.000	766.1	0.0000
RAG-only	0	0.5314	0.5185	1.000	955.1	0.0000
RAG-only	1	0.8358	0.8704	1.000	840.8	0.0000
RAG + neural loop	0	0.6890	0.6889	1.904	1511.5	0.0926
RAG + neural loop	1	0.9124	0.9630	1.681	1231.2	0.0556
RAG + symbolic loop	0	0.5314	0.5185	1.022	969.3	0.0000
RAG + symbolic loop	1	0.8176	0.8519	3.630	2409.4	0.5333
RAG + neural + symbolic feedback	0	0.7323	0.7333	1.889	1510.3	0.0630
RAG + neural + symbolic feedback	1	0.9123	0.9593	1.630	1215.3	0.0593
Intrinsic self-critique	0	0.6791	0.6815	3.000	3086.7	0.0000
Intrinsic self-critique	1	0.8809	0.9222	3.000	2757.8	0.0000
External-role self-critique	0	0.6109	0.6111	3.000	3085.4	0.0000
External-role self-critique	1	0.8806	0.9222	3.000	2757.6	0.0000
Gemma-4 judge loop	0	0.6286	0.6296	1.389	1339.2	0.0111
Gemma-4 judge loop	1	0.8450	0.8815	1.033	869.2	0.0000
Blind resampling	0	0.6485	0.6444	1.456	1385.5	0.0000
Blind resampling	1	0.8995	0.9407	1.326	1115.3	0.0000

The blinded subset points in the same direction: human majority accuracy is 0.92 at both levels, whereas Verifier-B records 0.60 at Level 0 and 0.90 at Level 1. This makes measurement mismatch a plausible contributor, not an identified cause. The length-matched analysis in Section 6.5 still retains a Level-0 deficit, and the human comparison contains only 50 items per level [37].

6.2.2 Paired Comparisons and the Neural–Symbolic Contrast

Table 6.2 gives the nine planned comparisons with zero-shot. All effects are positive, every paired-bootstrap interval excludes zero, and every McNemar test is significant [16]. The bootstrap p -values and Benjamini–Hochberg q -values are all $2/10001$, the resolution floor for 10,000 resamples [6]. Effect sizes, intervals and discordant counts therefore carry the useful comparative information.

Table 6.2 Paired effects relative to zero-shot

Condition	Δp_B	95% paired-bootstrap CI	Condition-only success	Zero-shot-only success	McNemar p
Static few-shot	+0.1451	[0.1001, 0.1897]	137	54	1.66×10^{-9}
RAG-only	+0.1182	[0.0753, 0.1618]	122	54	3.18×10^{-7}
RAG + neural loop	+0.2354	[0.1934, 0.2772]	167	28	2.70×10^{-25}
RAG + symbolic loop	+0.1091	[0.0649, 0.1524]	121	58	2.86×10^{-6}
RAG + neural + symbolic feedback	+0.2570	[0.2151, 0.2987]	176	26	1.46×10^{-28}
Intrinsic self-critique	+0.2147	[0.1711, 0.2584]	160	34	9.62×10^{-21}
External-role self-critique	+0.1804	[0.1381, 0.2231]	146	39	9.51×10^{-16}
Gemma-4 judge loop	+0.1715	[0.1282, 0.2149]	147	46	1.66×10^{-13}
Blind resampling	+0.2087	[0.1664, 0.2510]	154	33	6.91×10^{-20}

Each row contains 540 paired outputs. The two success columns are the discordant counts used by McNemar’s test. Figure 6.1 shows the corresponding probability effects and intervals; neither display ranks the active conditions.

The post-hoc neural-plus-symbolic versus neural-only contrast uses the same 540 pairs. Its target-probability difference is +0.02159, concentrated at Level 0 (+0.04328; Level 1: -0.00009). Binary accuracy differs by +0.02037, with 26 versus 15 discordant successes (McNemar $p = 0.11728$). Because this contrast was chosen after the main results were known, it remains exploratory and does not establish hybrid superiority.

6.3 Retry Dynamics and Proxy Divergence

Figure 6.2 separates the failure-selected attempt trajectories from the same-case changes in the Verifier-A-minus-Verifier-B gap. Its line styles identify the three loop conditions, while marker shapes distinguish the two verifiers.

For the neural loop, the paired Verifier-A-minus-Verifier-B gap widens by 0.182802 from attempt 1 to 2 ($n=147$ continuing cases) and by 0.114836 from attempt 2 to 3 ($n=67$). For the neural-plus-symbolic loop, the widening is 0.141481 ($n=147$) and 0.145979 ($n=58$). The symbolic-only loop differs: its gap changes by -0.042224 ($n=193$) and +0.001396 ($n=165$).

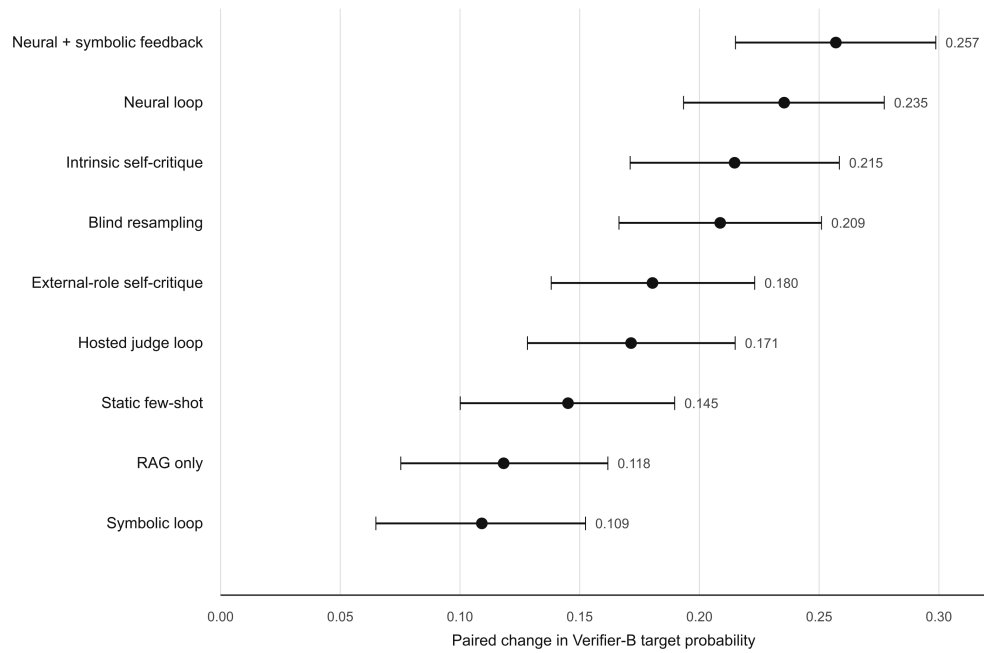


Fig. 6.1 Paired Verifier-B effects relative to zero-shot; whiskers show 95% bootstrap confidence intervals.

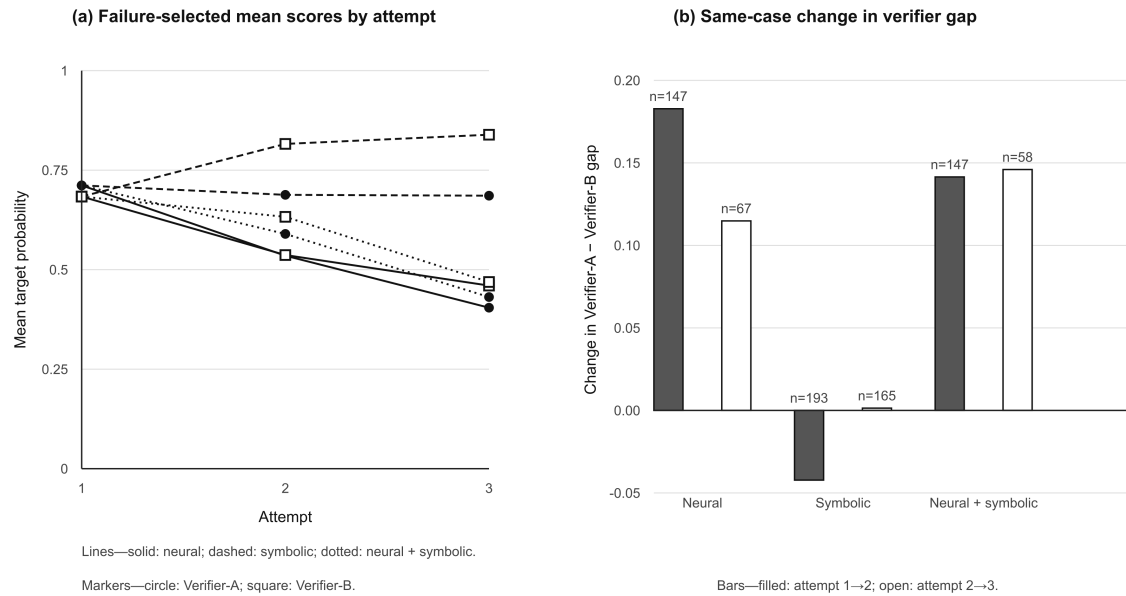


Fig. 6.2 Verifier trajectories and same-case gap changes among continuing failures.

Thus optimization against Verifier-A is associated with increasing A–B divergence in the two neural-gated loops, but not in the same form under the symbolic-only gate.

This is evidence of **measurable verifier divergence**, not proof that every revision is reward hacking. Later-attempt populations contain only previous failures, and Verifier-B’s own calibration improvement was not established. Recent evaluator-stress-test work likewise treats proxy–true divergence as a diagnostic rather than assuming the optimized evaluator remains valid [3]. Keeping Verifier-B outside the loop allows this divergence to be measured; it does not make Verifier-B an oracle.

6.4 Blinded Human Evaluation

Three native-Bangla annotators independently judged a frozen, balanced subset of 100 outputs under blinded condition labels. It contains five items from each condition-level cell and was sampled without verifier scores [67]. Persistent rater codes support the repeated-rating analysis [51, 46]. Table 6.3 reports all 300 judgments and panel agreement.

Table 6.3 Human recognition of the requested level

<i>Panel A. Annotator and pooled accuracy</i>				
Scope	n items	n judgments	Accuracy	95% case-bootstrap CI
Annotator A	100	100	0.9100	[0.8500, 0.9600]
Annotator B	100	100	0.9300	[0.8800, 0.9800]
Annotator C	100	100	0.9000	[0.8400, 0.9500]
Pooled judgments	100	300	0.9133	[0.8667, 0.9567]
<i>Panel B. Panel-level agreement</i>				
Agreement measure	Estimate	95% item-bootstrap CI	Interpretation	
Raw three-way agreement	0.8800	Not estimated	All three annotators selected the same level on 88 of 100 items	
Nominal Krippendorff alpha	0.8405	[0.7473, 0.9200]	Agreement beyond chance under the prespecified nominal coefficient	

Pooled accuracy is 0.9133, with 137/150 correct judgments at each level. Five Level-0 and seven Level-1 items split 2-to-1; all others are unanimous. The result supports recognition of the requested level on this subset, not condition-level performance, writing quality, factuality or audience response.

6.4.1 Human and Verifier Decisions on the Same Items

Table 6.4 compares the human majority label with both verifiers on the same 100 items. Majority voting yields one human label per item and an accuracy of 0.92; Table 6.3's 0.9133 instead pools all 300 judgments. This post-hoc comparison is exploratory, and Verifier-B on all 5,400 outputs remains the primary outcome.

Table 6.4 Human and verifier decisions on 100 blinded items

Panel A. Target-match accuracy and raw agreement

Stratum	n items	Human majority accuracy	Verifier-B accuracy	Verifier-A accuracy	Human-B raw agreement
Level 0	50	0.92	0.60	0.70	0.60
Level 1	50	0.92	0.90	0.84	0.86
Pooled	100	0.92	0.75	0.77	0.73

Panel B. Paired human-majority and Verifier-B disagreements

Stratum	Human-only correct	Verifier-B-only correct	Exact McNemar p
Level 0	18	2	0.000402
Level 1	4	3	1.0
Pooled	22	5	0.001514

Human majority accuracy is 0.92 at both levels. Verifier-B falls from 0.90 at Level 1 to 0.60 at Level 0, and Verifier-A from 0.84 to 0.70. At Level 0 the human panel alone is correct on 18 items, against two for Verifier-B alone; the Level-1 counts are four and three. The asymmetry is therefore present in the automatic instruments but not in the human majority labels on this subset.

Cohen's kappa is 0.46 pooled, 0.038 at Level 0 and 0.146 at Level 1. Within a single level the requested label is constant and the marginals are skewed, so raw agreement is more informative here [29]. With only 50 items per level, the comparison identifies a measurement concern but cannot decide which operationalization is closer to the construct [37].

6.5 Length-Matched Sensitivity Analysis

The preregistered slice pairs Level-0 and Level-1 outputs from the same plot, condition and replicate when their word counts differ by less than 15%. It retains 486 of 2,700 possible pairs. Table 6.5 reports accuracy together with coverage because conditioning on generated length is post-treatment selection.

Table 6.5 Outcomes in the length-matched sensitivity slice

Condition	Matched pairs / 270	Coverage	Mean absolute word gap	Accuracy all	L0	L1
Zero-shot	30	11.11%	1.267	0.6333	0.4667	0.8000
Static few-shot	40	14.81%	1.100	0.7625	0.7250	0.8000
RAG-only	70	25.93%	1.157	0.7500	0.5857	0.9143
RAG + neural loop	64	23.70%	1.156	0.8359	0.7031	0.9688
RAG + symbolic loop	67	24.81%	1.224	0.7537	0.6119	0.8955
RAG + neural + symbolic feedback	57	21.11%	1.105	0.8684	0.7544	0.9825
Intrinsic self-critique	11	4.07%	0.818	0.9545	1.0000	0.9091
External-role self-critique	9	3.33%	1.111	0.9444	1.0000	0.8889
Gemma-4 judge loop	58	21.48%	1.155	0.8448	0.7241	0.9655
Blind resampling	80	29.63%	1.088	0.8188	0.6875	0.9500

Level-0 accuracy remains lower in eight of ten conditions. The two exceptions, intrinsic and external-role self-critique, retain only 11 and 9 pairs. Length therefore does not explain the full Level-0 deficit, but the uneven, post-treatment coverage prevents a length-neutral control claim.

6.6 Diversity and Distributional Diagnostics

Figure 6.3 keeps three corpus-level diagnostics separate.

Jensen–Shannon word-count divergence ranges from 0.153390 to 0.611987. In nine conditions, Level 1 lies further from its level-specific real-review reference; external-role self-critique is the exception. Very short outputs are concentrated in the two Level-0 self-critique cells: 42.59% for external-role and 31.11% for intrinsic critique, compared with at most four cases in each remaining cell. Lexical diversity is reported separately rather than folded into realism. Across the 20 cells, Distinct-1 ranges from 0.210526 to 0.332641, Distinct-2 from 0.499618 to 0.728167, and Self-BLEU-4 from 0.137042 to 0.477088. The lowest Self-BLEU-4 occurs for external-role self-critique at Level 1; the highest occurs for zero-shot at Level 0. These ratios are length-sensitive corpus diagnostics and do not constitute a quality ranking.

LaBSE-feature MAUVE ranges from 0.010463 to 0.035995. With 270 generated and 270 real texts per cell, it is a small-sample sensitivity analysis and is not comparable with default GPT-2/MoP MAUVE [47]. Sentiment divergence remains unmeasured because no independent generated-text sentiment scorer exists.

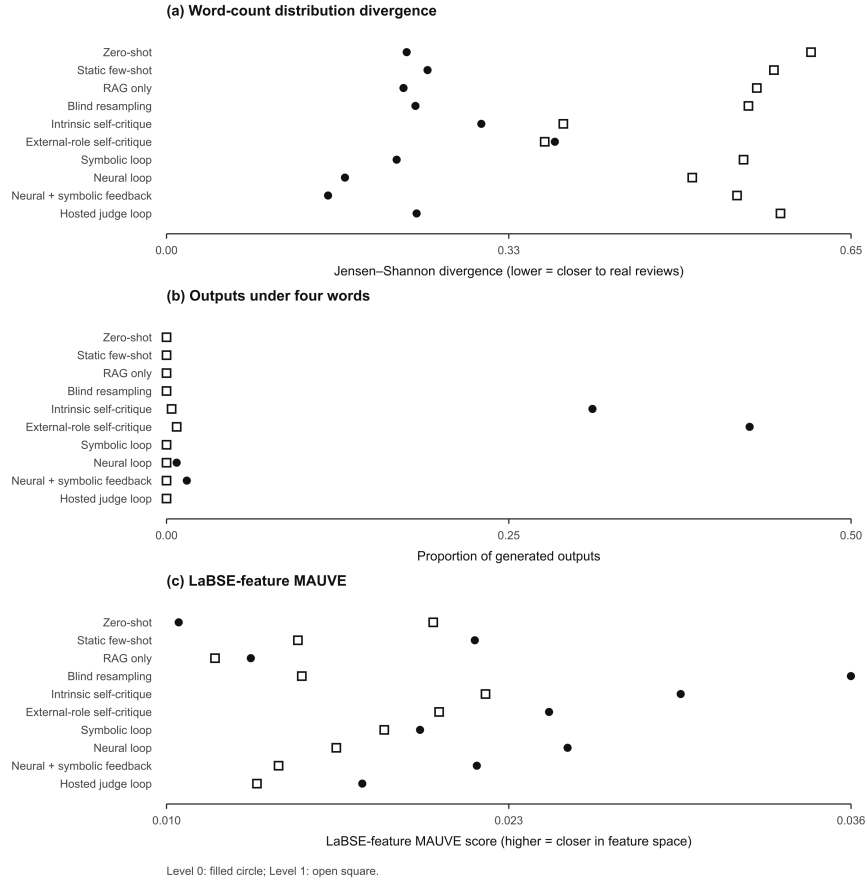


Fig. 6.3 Length and feature-space diagnostics across conditions.

6.7 Qualitative Error Analysis

Six strata were fixed before reading the outputs; the first seed-42 case key in each stratum was selected. Table 6.6 records the selection and measurements, and Figure 6.4 shows the six distinct responses. They are illustrations of the registered outcome, not human-gold judgments.

E5' and E1 are the same baseline case, so Figure 6.4 omits E5'.

E1 is a short baseline failure accepted by neither verifier. E3 lies near Verifier-B's boundary but is strongly accepted by Verifier-A; E2 and E4 show the same directional disagreement at Level 1. E5 is the treatment member of a same-case repair pair: target probability rises from 0.1638 to 0.9853 as length increases from seven to fourteen words.

E6 exposes the symbolic fallback. Across its three drafts, p_B is 0.9414, 0.1173 and 0.7380, yet the controller emits the second draft selected by its gate score. At seed 42, 50/90 Level-1 cases exhaust the budget and 6/90 also fail Verifier-B. Budget exhaustion and outcome failure are therefore related but not equivalent.

Table 6.6 Selection basis for the reported examples, seed 42

<i>Panel A. Selection metadata</i>				
ID	Condition	Level	Selection stratum	Stratum size / 90
E1	Zero-shot	0	Verifier-B failure	53
E2	Zero-shot	1	Verifier-B failure	19
E3	RAG + neural + symbolic feedback	0	Residual failure	27
E4	RAG + neural + symbolic feedback	1	Residual failure	3
E5	RAG + neural + symbolic feedback	0	Repaired pair, treatment	36
E5'	Zero-shot	0	Repaired pair, baseline	—
E6	RAG + symbolic loop	1	Failure after budget exhaustion	6
<i>Panel B. Text and verifier measurements</i>				
ID	Words	Verifier-B target probability	Verifier-A target probability	Model calls
E1	7	0.1638	0.0012	1
E2	20	0.4147	0.7117	1
E3	14	0.4894	0.9925	1
E4	15	0.0567	0.9850	1
E5	14	0.9853	1.0000	1
E5'	7	0.1638	0.0012	1
E6	24	0.1173	0.0000015	5

6.8 Scope of the Comparisons

The nine internal comparators share the task, plots and scoring contract, which supports paired comparisons with zero-shot. Blind resampling is matched on the admitted model-token budget, not physical runtime, FLOPs or latency. No external system is evaluated. The Chapter 2 review identified no Bangla system with the same engagement-specificity task and sealed outcome verifier, and the corpus lacks film identifiers needed for film-matched references. An external comparison would therefore require a new, task-matched evaluation.

6.9 Chapter Summary

The Bangla results support control of the requested level, but automatic measurement is weaker at Level 0 and the two verifiers diverge during neural-gated retries. Chapter 7

Example	Selection role	Generated Bangla response
E1	Baseline failure (L0)	ব্যাপারটা খুবই দুঃখজনক! এমন একটা গল্প! 😞😞
E2	Baseline failure (L1)	গল্পটা দেখাচ্ছিল, একটা মেয়ের জীবন এভাবে শেষ হয়ে যায়! মেনে নিতে পারলাম না, এত সুন্দর একটা প্রাণ এভাবে চলে গেল।
E3	Residual failure (L0)	ব্যাপারটা পুরাই অন্যরকম! শেষটা মনে হল যেন একটু বেশি প্যাঁচানো ছিল, তবে ছবিটা ভালোই।
E4	Residual failure (L1)	আশা মারা গেল এটা মেনে নিতে পারিনি, এমন একটা সুন্দর গল্পে এত বড় একটা ধাক্কা!
E5	Repaired output (L0)	অসাধারণ! গল্পটা খুবই হৃদয়স্পর্শী, এমন একটা ছবি দেখে মনটা একেবারে খাঁ খাঁ করে উঠলো।
E6	Budget exhaustion (L1)	ছবিটা দেখেই বোঝা যাচ্ছে মাল কাহিনীর চেয়ে বেশি কিছু না। নায়কের অ্যাকশনগুলো একটু বেশিই ভেলান্টিয়ার লাগছে, আর ফুল আর ধনী বাবুর প্রেম—এটা আগে দেখিনি!

Fig. 6.4 Rule-selected Bangla responses.

interprets these findings; the supplementary appendix reports the complete per-seed and diagnostic results.

Chapter 7

Discussion and Limitations

Chapter 6 showed that generated Bangla cinema responses can be made more consistent with a requested engagement-specificity level, while also exposing a gap between successful control and faithful audience modelling. This chapter interprets that gap and considers the limits of the evidence. Contributions and future work are reserved for the final chapter.

7.1 Main Finding

The evidence shows that conditioning and revision make short Bangla cinema responses more consistent with a requested engagement-specificity level than zero-shot prompting. Verifier-B provides consistent coverage of the complete frozen generation surface, and native Bangla readers usually identify the same distinction on a smaller blinded subset. Neither result demonstrates that the system reproduces a real audience. The generated responses were not matched against reactions to the same films, and the two levels do not represent demographic or psychological groups. The result is therefore about control over a textual distinction, not prediction of viewers, sentiment, popularity, or commercial performance. This boundary is especially important because synthetic-audience research is vulnerable to prompt sensitivity, hallucination and anthropomorphic interpretation [35]. No single mechanism accounts for the improvement: static examples, retrieval, blind resampling, self-critique and verifier-guided revision all improve on zero-shot under the planned comparisons. The neuro-symbolic condition has the largest observed difference from zero-shot, but the experiment does not rank the active systems. Its direct comparison with the neural-only loop was post-hoc and remains exploratory.

7.1.1 Engagement Specificity as a Continuum

The early corpus analysis changed the meaning of the target variable. The Region-A partition was reproducible, yet the geometric evidence did not support two separated clusters: the silhouette was weak, the gap statistic selected no value of K , and HDBSCAN treated the points as noise. Human comparative judgment showed that readers could nevertheless recognize the contrast and its direction. This combination supports a usable cut through a continuum, not two naturally occurring audience types. The framework can therefore request a broader or more specific response without assigning a person to a fixed category. It controls a property of the generated text; it does not infer personas within Bangla-speaking audiences.

Length remains part of the construct problem. The corpus, development outputs and final generations all show that engagement specificity and response length are related, although the direction is not stable across every setting. The length-matched analysis retains a Level-0 deficit, so length alone does not explain the result. At the same time, matching keeps only a selective fraction of the outputs and cannot establish length-neutral control. The model appears to manipulate a bundle of cues—detail, reference to plot events, evaluative specificity and length—rather than an isolated semantic property.

7.2 What the Control Mechanisms Add

The ablations separate four functions that would otherwise be hidden inside one system score:

- **Examples and retrieval.** Static demonstrations and R1 retrieval show the two levels in Bangla. They improve the prompt but do not decide when a draft has reached its target.
- **Neural verification.** Verifier-A supplies the stopping decision. Revision costs additional calls, and later attempts contain only earlier failures, so their trajectory describes difficult survivors rather than a fixed population.
- **Symbolic diagnosis.** Symbolic acceptance is weak and inefficient, particularly at Level 1. Symbolic feedback is more useful for naming a problem while the neural verifier retains the stopping decision. The Level-0 increment over neural-only is exploratory, not evidence of general hybrid superiority.
- **Alternative critics.** Self-critique improves control at a fixed three-call cost. The hosted judge is a comparator, not an independent outcome authority. Evaluator replacement, self-preference and uneven multilingual agreement reinforce this restriction [65, 48, 19, 49].

7.2.1 Why Verifier Separation Matters

Verifier separation becomes most informative during revision. In the neural-gated loops, drafts selected to satisfy Verifier-A receive progressively less comparable support from Verifier-B. Because both scores are available on the same continuing cases, the divergence cannot be attributed to a change in plot or requested level. Improvement against the in-loop proxy and improvement under a held-out instrument are not interchangeable.

This pattern is consistent with Goodhart-style pressure, but the evidence does not identify reward hacking in every revised response. Later attempts are failure-selected, Verifier-B has its own errors, and its calibration improvement was not established. Repeated optimization nevertheless changes the relationship between the two instruments in the direction expected when a proxy becomes increasingly exploitable. Evaluator stress-testing work similarly argues that an optimized score needs independent or invariant checks before it can stand for the intended construct [3]. Keeping Verifier-B outside generation therefore does more than prevent direct leakage. It makes disagreement observable. Had the same verifier controlled revision and supplied the final outcome, the experiment could report rising scores without showing whether the revisions generalized to another trained instrument. The separation does not make Verifier-B ground truth, but it turns proxy dependence into an empirical question.

7.3 Human and Automatic Evidence

The blinded study tests the meaning of the output labels rather than providing another condition leaderboard. Native Bangla readers usually recover the requested level with strong panel agreement, but five items per condition-level cell are too few for reliable system ranking. The same-item comparison reveals an important asymmetry. Human majority labels perform similarly at both levels, whereas both automatic verifiers are weaker at Level 0. This makes an instrument-specific Level-0 problem plausible. It does not prove that the human panel is the definitive oracle: the comparison contains only 100 items, and within-level agreement statistics are affected by fixed requested-label marginals [37, 29]. Still, the discrepancy warns against describing Verifier-B accuracy as human quality. Annotators selected the requested level; they did not rate fluency, plot faithfulness, sentiment, usefulness, offensiveness or resemblance to real audience responses. Human validation therefore supports controllability of the stated distinction, not overall response quality.

7.4 Practical Implications

The evidence supports three practical decisions:

- **Appropriate use:** an inspectable pre-writing aid for comparing both levels, reviewing retrieved examples and revision status, and preparing questions for direct audience research.
- **Unsupported use:** autonomous marketing, casting, segmentation or commercial forecasting. The study learns no film-level response distribution and links no generated statement to a demographic group.
- **Deployment choice:** static prompting, retrieval or blind resampling may suit inexpensive variety; neural-guided loops add stopping and correction traces. The symbolic-only loop is not supported as a default, and cost should not be hidden inside one quality score.

7.5 Limitations

7.5.1 Construct and Measurement

- The axis comes from one Bangla review corpus with unresolved source history; region, sentiment and length complicate its geometric origin. Human validation supports an operational label, not a natural audience taxonomy.
- Verifier-B is a fixed instrument, not a calibrated probability of human acceptance. Its Level-0 mismatch and calibration null remain unresolved.
- The realism measures capture different properties. LaBSE-feature MAUVE is small-sample sensitivity evidence [47], and sentiment divergence remains unmeasured because no independent registered scorer was available.

7.5.2 Internal and Statistical Validity

- Verifier-A and Verifier-B use disjoint data and different model families, but learn the same operational label; their errors need not be independent. The hosted judge is also a same-family treatment, not an external authority.
- Later attempts are failure-selected, and length matching conditions on a property changed by generation. Neither analysis supports an unrestricted causal reading.
- The primary tests use 540 paired cases, paired bootstrap, McNemar testing and Benjamini–Hochberg correction [16, 6]. The three seeds are sensitivity blocks, not

independent replications, and the direct hybrid contrast remains outside the registered comparison family.

7.5.3 External Validity, Human Evaluation, and Ethics

- The corpus has no film identifier. The 90 held-out plots separate evaluation from development but do not establish generalization to other Bangla registers, communities, domains or generator families.
- The three adult native-Bangla evaluators were university batchmates known to the researcher. Blinding and coded identities do not remove the limits of a convenience sample or possible social pressure. No institutional approval or exemption is claimed, and absent fields were not reconstructed [5].
- Outputs may contain stereotypes, offensive language or unsupported plot details. The interface's source-support diagnostic was not part of the frozen experiment or a registered human faithfulness audit; it neither establishes a hallucination rate nor certifies safety.

7.6 Chapter Summary

The study shows that engagement specificity can be varied in short Bangla cinema responses, while the verifier and human evidence explains why this result is not audience simulation. Chapter 8 closes the argument and identifies the evidence still needed beyond this corpus.

Chapter 8

Conclusion and Future Work

This chapter draws together the findings from the Bangla study and outlines the questions that remain. No new analysis is introduced.

8.1 Conclusion

This thesis examined whether a verifier-in-the-loop workflow could control a meaningful distinction in short Bangla cinema responses while revealing the risks of optimizing an automatic proxy. The Bangla experiment shows that the registered controls improve Verifier-B target probability over zero-shot generation, and native Bangla readers usually recover the requested level on the blinded subset. That level represents engagement specificity, not audience identity. The corpus does not contain natural audience clusters, response length remains entangled with the two levels, and the observed advantage of symbolic feedback over neural-only gating remains exploratory. Symbolic rules are therefore better supported as revision diagnostics than as an acceptance gate. Because Verifier-B remains outside generation, its widening disagreement with Verifier-A can be observed rather than optimized away. The framework supports controlled Bangla response generation, but it does not model real audiences. Claims about viewers, films or market response still require evidence from people.

8.2 Future Work

Several questions remain for future research:

- **Clarify what symbolic feedback adds.** A preregistered comparison with neural-only gating should define level-wise contrasts before generation, retain paired cases and keep the outcome verifier sealed.
- **Extend the human evaluation.** A larger and more diverse participant pool is needed for condition-level estimates. Target match, fluency, naturalness, usefulness and harmful content should remain separate measures rather than being merged into one score.
- **Check plot faithfulness.** A blinded human audit should compare each response with its source plot and record unsupported or contradictory details. The interface diagnostic should be checked against these judgments before it is treated as an evaluation measure.
- **Probe the verifiers.** Controlled changes in length, register, punctuation, sentiment markers and lexical specificity could show which cues drive disagreement between Verifier-A and Verifier-B. Calibration also needs a larger held-out set.
- **Rebuild symbolic verification before using it as a gate.** A revised scorer would need more labelled data and out-of-sample evaluation. Until then, its supported role is diagnostic.
- **Test beyond the present setting.** The deferred English mirror, broader Bangla registers, longer responses and other generator families would show which findings travel beyond this study. Film-linked human responses are necessary before controlled generation can be considered for pre-release audience prediction.

References

- [1] Adib, S. E. S., Sani, A. A., Esham, E. A., Abrar, A., Tashdeed, I., and Chowdhury, M. T. A. (2026). Benhallueval: A multi-task hallucination evaluation framework for large language models on bengali. *arXiv preprint arXiv:2605.31483*.
- [2] Aich, R., Das, N., and Chowdhury, M. (2026). HybridRAG-BN: A retrieval-augmented framework with fine-tuned verification for bangla KBQA.
- [3] Akter, S., Shihab, I. F., and Sharma, A. (2026). Detecting proxy gaming in RL and LLM alignment via evaluator stress tests. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 10554–10583.
- [4] Balanya, S. A., Maro nas, J., and Ramos, D. (2022). Adaptive temperature scaling for robust calibration of deep neural networks. *Neural Computing and Applications*.
- [5] Belz, A. and Thomson, C. (2025). HEDS 3.0: The human evaluation data sheet version 3.0. In *Proceedings of the Fourth Workshop on Generation, Evaluation and Metrics*, pages 60–81.
- [6] Benjamini, Y. and Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300.
- [7] Bhattacharjee, A., Hasan, T., Ahmad, W. U., Samin, K., Islam, M. S., Iqbal, A., Rahman, M. S., and Shahriyar, R. (2022). BanglaBERT: Language model pretraining and benchmarks for low-resource language understanding evaluation in bangla. In *Findings of the Association for Computational Linguistics: NAACL 2022*, pages 1318–1327.
- [8] Bilal, A., Mohsin, M. A., Umer, M., Trigg, L., Subhan, A., Ali, M., and Hougen, D. F. (2026). Refining over resampling: Test-time self-correction for LLM reasoning.
- [9] Bucher, M. J. J. and Martini, M. (2024). Fine-tuned ’small’ LLMs (still) significantly outperform zero-shot generative AI models in text classification. *arXiv preprint*.
- [10] Bui, N., Nguyen, H. T., Kumar, S., Theodore, J., Qiu, W., Nguyen, V. A., and Ying, R. (2025). Mixture-of-personas language models for population simulation.
- [11] Campello, R. J. G. B., Moulavi, D., and Sander, J. (2013). Density-based clustering based on hierarchical density estimates. In *Advances in Knowledge Discovery and Data Mining (PAKDD)*.
- [12] Casola, S., Lauriola, I., and Lavelli, A. (2022). Pre-trained transformers: an empirical comparison. *Machine Learning with Applications*.

- [13] Chen, K.-Y., Su, F.-Y., Lin, S.-Y., Li, B., and Chiang, J.-H. (2026). The self-correction illusion: Role relabeling gates explicit error flagging in large language models.
- [14] Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Plappert, M., Tworek, J., Hilton, J., Nakano, R., Hesse, C., and Schulman, J. (2021). Training verifiers to solve math word problems.
- [15] Cui, W., Zhang, H., Huang, Y., Li, Q., Xu, J., Liu, C.-L., Gao, C., Wang, J., and Zhang, B. (2026). SymDiag: Explainable diagnosis for LLM reasoning via neuro-symbolic verification. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*.
- [16] Dror, R., Baumer, G., Shlomov, S., and Reichart, R. (2018). The hitchhiker’s guide to testing statistical significance in natural language processing. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1383–1392.
- [17] Fageot, J., Grossglauser, M., Hoang, L.-N., Tacchi-Bénard, M., and Villemaud, O. (2026). The benefits of diversity: Combining comparisons and ratings for efficient scoring.
- [18] Feng, F., Yang, Y., Cer, D., Arivazhagan, N., and Wang, W. (2022). Language-agnostic BERT sentence embedding. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 878–891.
- [19] Fu, X. and Liu, W. (2025). How reliable is multilingual LLM-as-a-judge? In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 11040–11053.
- [20] Fu, Z., So, A. M.-C., and Collier, N. (2023). A stability analysis of fine-tuning a pre-trained model.
- [21] Gundersen, O. E., Shamsaliei, S., Kjærnli, H. S., and Langseth, H. (2023). On reporting robust and trustworthy conclusions from model comparison studies involving neural networks and randomness. In *Proceedings of the 2023 ACM Conference on Reproducibility and Replicability*.
- [22] Guo, C., Pleiss, G., Sun, Y., and Weinberger, K. Q. (2017). On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70, pages 1321–1330.
- [23] Gwet, K. L. (2008). Computing inter-rater reliability and its variance in the presence of high agreement. *British Journal of Mathematical and Statistical Psychology*, 61(1):29–48.
- [24] Hassin, F., Rakin, A., Alom, M. S., Ahamed, M. I., Mondal, S., and Islam, A. (2026). Banglablend based evaluation of Bengali formal and colloquial text classification. In *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*.
- [25] Hossain, S. (2026). A neuro-symbolic multi-agent framework for pre-release audience response generation in bangla cinema: A verifier-in-the-loop approach. GitHub repository. Source code and supplementary materials. Accessed: Sep. 2, 2026.

- [26] Hossain, S., Islam, H., Khan, N., Alam, M. M., Uddin, M. J., and Islam, M. R. (2026). Raw bangla movie review comment dataset for sentiment analysis and natural language processing. Mendeley Data, Version 3.
- [27] Huang, J., Chen, X., Mishra, S., Zheng, H. S., Yu, A. W., Song, X., and Zhou, D. (2024). Large language models cannot self-correct reasoning yet. In *The Twelfth International Conference on Learning Representations*.
- [28] Hubert, L. and Arabie, P. (1985). Comparing partitions. *Journal of Classification*, 2:193–218.
- [29] James, J. (2026). Counting on consensus: Selecting the right inter-annotator agreement metric for NLP annotation and evaluation. *arXiv preprint arXiv:2603.06865*.
- [30] Kamoi, R., Zhang, Y., Zhang, N., Han, J., and Zhang, R. (2024). When can LLMs actually correct their own mistakes? a critical survey of self-correction of LLMs. *Transactions of the Association for Computational Linguistics*, 12:1417–1440.
- [31] Kapur, R., Hawkins, R., and Kreiss, E. (2026). When more words say less: Decoupling length and specificity in image description evaluation. *arXiv preprint arXiv:2601.04609*.
- [32] Kiritchenko, S. and Mohammad, S. M. (2017). Best-worst scaling more reliable than rating scales: A case study on sentiment intensity annotation. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 465–470.
- [33] Kotte, V. (2026). Ucci: Calibrated uncertainty for cost-optimal llm cascade routing. *arXiv preprint arXiv:2605.18796*.
- [34] Krippendorff, K. (2018). *Content Analysis: An Introduction to Its Methodology*. SAGE Publications, Thousand Oaks, CA, USA, 4 edition.
- [35] Lappas, T. and Filippas, A. (2026). Synthetic audiences as proxies for human users: A systematic literature review. *IEEE Access*, 14:94899–94932.
- [36] Laurer, M., van Atteveldt, W., Casas, A., and Welbers, K. (2023). Less annotating, more classifying: Addressing the data scarcity issue of supervised machine learning with deep transfer learning and BERT-NLI. *Political Analysis*.
- [37] Levtssov, G. and Ustalov, D. (2025). Confidence and stability of global and pairwise scores in NLP evaluation. *arXiv preprint arXiv:2507.01633*.
- [38] Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., tau Yih, W., Rocktäschel, T., Riedel, S., and Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474.
- [39] Liu, A. and Meng, J. (2026). Self-correction as feedback control: Error dynamics, stability thresholds, and prompt interventions in LLMs.

- [40] Madaan, A., Tandon, N., Gupta, P., Hallinan, S., Gao, L., Wiegrefe, S., Alon, U., Dziri, N., Prabhumoye, S., Yang, Y., Gupta, S., Majumder, B. P., Hermann, K., Welleck, S., Yazdanbakhsh, A., and Clark, P. (2023). Self-refine: Iterative refinement with self-feedback.
- [41] Mahmoud, A., Rezaei, M., Wang, Z., Gunjal, A., Liu, B., and He, Y. (2026). Reward hacking in rubric-based reinforcement learning. *arXiv preprint arXiv:2605.12474*.
- [42] Mazumder, S. K., Jannat, S., and Mondal, S. (2025). Transformer-based approach for accurately identifying saint and common forms of Bangla language. In *2025 IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE)*.
- [43] McInnes, L., Healy, J., and Astels, S. (2017). hdbscan: Hierarchical density based clustering. In *The Journal of Open Source Software*, volume 2, page 205.
- [44] Mirzaei, I. (2026). Sample more, reflect less: Self-refine and reflexion lose to repeated sampling at equal token cost, from 1.5b to 7b.
- [45] Mitra, S. K., Riyad, T., Hossan, M. F., Sen, P., Islam, S. J., and Shara, S. T. (2025). Exploring MuRIL for emotion detection in Bangla text. In *2025 IEEE 4th International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON)*.
- [46] Pandita, D., Korn, F., Welty, C. A., and Homan, C. (2026). Improving reproducibility in evaluation through multi-level annotator modeling. *arXiv preprint arXiv:2605.13801*.
- [47] Pillutla, K., Swayamdipta, S., Zellers, R., Thickstun, J., Welleck, S., Choi, Y., and Harchaoui, Z. (2021). MAUVE: Measuring the gap between neural text and human text using divergence frontiers. In *Advances in Neural Information Processing Systems*, volume 34, pages 4816–4828.
- [48] Pombal, J., Rei, R., and Martins, A. F. T. (2026). Self-preference bias in rubric-based evaluation of large language models. In *Conference on Language Modeling (COLM)*.
- [49] Ponce, D., Gete, H., Etchegoyhen, T., Zubiaga, I., and Soroa, A. (2026). Judging instruction responses in a low-resource language: A case study on basque. In *Proceedings of the Fifteenth Language Resources and Evaluation Conference*, pages 281–298.
- [50] Rieder, T., Schneider, M., Truss, M., Tsaplin, V., Rublea, A., Dere, S., Sanz, F. C., Reiss, T., and Dogan, M. D. (2026). SimAB: Simulating A/B tests with persona-conditioned AI agents for rapid design evaluation.
- [51] Riley, P., Deutsch, D., Foster, G. F., Ratnakar, V., Dabirmoghaddam, A., and Freitag, M. (2024). Finding replicable human evaluations via stable ranking probability. In *Proceedings of NAACL-HLT 2024*.
- [52] Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20:53–65.

- [53] Roy, A., Rahaman, R., Shibly, S., Saha, U., Kafi, A. A., and Sadeque, F. Y. (2025). Advancing bangla machine translation through informal datasets. *arXiv preprint arXiv:2512.13487*.
- [54] Saleh, A., Tesfay, T. W., Nguyen, P. T., Rocco, J. D., Zeshan, M. U., and Ruscio, D. D. (2026). The illusion of agentic complexity in README.md generation: Evaluating single-agent vs. multi-agent RAG systems.
- [55] Sands, B., Wang, Y., Xu, C., Zhou, Y., Wei, L., and Chandra, R. (2026). An evaluation of LLMs for generating movie reviews: GPT-4o, Gemini-2.0 and DeepSeek-V3. *Neural Computing and Applications*, 38(13):556.
- [56] Schneider, L., Bischl, B., and Feurer, M. (2025). Overtuning in hyperparameter optimization. In *AutoML 2025*.
- [57] Shinn, N., Cassano, F., Berman, E., Gopinath, A., Narasimhan, K., and Yao, S. (2023). Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 36.
- [58] Shuvo, M. A., Hasan, M., Parvez, M. T., Noman, A. H., and Ovi, M. S. H. (2026). Polite on the surface, wrong in practice: A curated dataset for fixing honorific failures in multilingual bangla generation. *arXiv preprint arXiv:2605.22487*.
- [59] Sijan, T. A., Rifat, S. M. G., Partha, P. C., Islam, M. T., and Anwar, M. M. (2026). Banglasocialbench: A benchmark for evaluating sociopragmatic and cultural alignment of llms in bangladeshi social interaction. *arXiv preprint arXiv:2603.15949*.
- [60] Tibshirani, R. and Walther, G. (2005). Cluster validation by prediction strength. *Journal of Computational and Graphical Statistics*, 14(3):511–528.
- [61] Tibshirani, R., Walther, G., and Hastie, T. (2001). Estimating the number of clusters in a data set via the gap statistic. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 63(2):411–423.
- [62] Tunstall, L., Reimers, N., Jo, U. E. S., Bates, L., Korat, D., Wasserblat, M., and Pereg, O. (2022). Efficient few-shot learning without prompts. *arXiv preprint*.
- [63] Vach, W. and Gerke, O. (2023). Gwet’s AC1 is not a substitute for Cohen’s kappa – a comparison of basic properties. *MethodsX*.
- [64] Yang, K. and Klein, D. (2021). FUDGE: Controlled text generation with future discriminators. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 3511–3535.
- [65] Yang, Z., Hou, Y., and Yang, X. (2026). When the judge changes, so does the measurement: Auditing LLM-as-judge reliability.
- [66] Zhang, J., Zhong, L., Zhu, Z., Huang, S., and Xu, Y. (2026). TabPFN beyond tabular data: Calibration and accuracy on multimodal embeddings. *arXiv preprint arXiv:2607.11007*.
- [67] Zouhar, V., Cui, P., and Sachan, M. (2025). How to select datapoints for efficient human evaluation of NLG models? *Transactions of the Association for Computational Linguistics*, 13:1789–1811.